

Cpis499 Senior Project

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This report is submitted in partial fulfillment of the requirements for the award of Bachelor Science in Information Systems

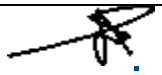
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Declaration of Originality

We hereby declare that this project report is based on our original work except for citations and quotations, which are duly acknowledged. We also declare that this project has not been previously and concurrently submitted for any other degree or award at KAU or other institutions.

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Abstract

The digital transformation of Saudi Arabia under Vision 2030 focuses on artificial intelligence and data-driven technologies to boost different sectors throughout the country. The analysis of public sentiments using AI technology stands as a primary area of attention because digital platforms and online reviews have increased its importance. The proposed system develops AI technology through the integration of NLP and ML to automate product review classification into positive, negative and neutral categories. The system uses Aspect-Based Sentiment Analysis (ABSA) to determine key product features while evaluating their positive or negative sentiments. The proposed system helps businesses discover valuable customer feedback which enables them to enhance their products and services. The system enables consumers to base their purchase decisions on precise sentiment analysis results. The tool ensures high accuracy as well as real-time processing for sentiment classification and review summarization tasks through its efficient and scalable approach to address traditional sentiment analysis limitations.

الخلاصة

يركز التحول الرقمي في المملكة العربية السعودية في ظل رؤية ٢٠٣٠ على الذكاء الاصطناعي والتقنيات القائمة على البيانات لتعزيز القطاعات المختلفة في جميع أنحاء البلاد. ويشكل تحليل المشاعر العامة باستخدام تقنية الذكاء الاصطناعي مجالًا أساسيًا للاهتمام لأن المنصات الرقمية والمراجعات عبر الإنترنت زادت من أهميتها.

يطور النظام المقترح تقنية الذكاء الاصطناعي من خلال دمج معالجة اللغة الطبيعية والتعلم الآلي لأتمتة تصنيف مراجعات المنتج إلى فئات إيجابية وسلبية ومحايدة. يستخدم النظام تحليل المشاعر القائم على الجوانب (ABSA) لتحديد ميزات المنتج الرئيسية أثناء تقييم مشاعرهم الإيجابية أو السلبية.

يساعد النظام المقترح الشركات على اكتشاف ملاحظات العملاء القيمة التي تمكنهم من تحسين منتجاتهم وخدماتهم. يتيح النظام للمستهلكين الاستناد إلى قرارات الشراء الخاصة بهم على نتائج تحليل المشاعر الدقيقة. تضمن الأداة دقة عالية بالإضافة إلى المعالجة في الوقت الفعلي لتصنيف المشاعر ومهام تلخيص المراجعة من خلال نهجها الفعال والقابل للتطوير لمعالجة قيود تحليل المشاعر التقليدية.

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Chapter 1

1.1Motivation and Context

The rapid growth of internet-based applications, such as social media platforms and blogs, has resulted in an increase in comments and reviews related to daily activities. Sentiment analysis is the process of collecting and analyzing individuals' opinions, ideas, and impressions on various topics, products, and services. These opinions can be invaluable to businesses, governments, and individuals as they gather information and make decisions based on public sentiment. However, the process of analyzing and evaluating emotions faces numerous challenges that hinder the accurate interpretation of feelings and the determination of the appropriate emotional polarity. In recent years, sentiment analysis has gained widespread acceptance, not only among researchers but also among companies, governments, organizations, and even famous. The growing popularity of the Internet has established it as the primary source of global information. By utilizing various online resources to express their opinions, consumers find that these platforms assist them in deciding whether to purchase a particular product.

Our project aims to tackle these challenges by developing a sentiment analysis tool for product reviews that utilizes advanced natural language processing (NLP) and machine learning (ML) technologies. This tool will offer a comprehensive product review rating system that categorizes sentiments as positive, negative, or neutral. Furthermore, it will conduct aspect-based sentiment analysis (ABSA) to identify product features that are mentioned in a positive or negative light.

1.2 Problem Statement

With the growth of digital platforms, businesses and content producers receive a substantial volume of audience feedback in the form of messages, reviews, and comments. However, most sentiment analysis systems currently in use rely on manual evaluation or basic tools that struggle to process unstructured and heterogeneous textual input. The complexity of language, which encompasses slang, sarcasm, and an evolving lexicon, presents significant challenges. Consequently, advanced sentiment analysis techniques are urgently needed to help organizations and content providers better analyze audience engagement, enhance customer interactions, and refine content strategy.

1.3 Project Aims and Objectives:

1.Help customers make informed decisions Provide clear and objective analysis of reviews, enabling customers to choose products or services that best suit their needs.

2.Improve products and services Help companies understand customer feedback better, allowing them to refine their offerings and enhance their overall experience. Empowering content creators craft more engaging and relevant content: By understanding audience feedback and trends, creators can tailor their content to better connect with their followers and drive meaningful engagement.

1.4 Scope and Assumptions

This project focuses on developing a tool for analyzing sentiment in product reviews shared on social media, helping users understand public opinions on various products. The project scope includes:

- AI-powered sentiment analysis to classify reviews as positive, negative, or neutral.
- Aspect-Based Sentiment Analysis (ABSA) to identify frequently mentioned product features and determine the sentiment associated with them.
- An interactive user interface that allows users to input text or upload reviews for instant analysis.
- Analytical reports displaying key insights, including commonly used keywords and sentiment trends to help users make informed decisions.
- Support for English-language reviews, with sentiment analysis initially available only for English text.

1.5 Stakeholders

Table 1-1: Stakeholders

Stakeholders	Description
Project Team	The developers, together with data scientists, hold responsibility for both developing and sustaining the sentiment analysis tool.
Evaluators Team	Experts in natural language processing and machine learning who assess the accuracy and performance of the system.
End Users	- Business owners, including e-commerce store owners, looking to analyze customer sentiment.
	- Consumers who want quick insights into product reviews before making a purchase.
	-Researchers and analysts studying public opinion trends.
	- Content creators looking to understand audience sentiment on their posts and videos.

1.6 Report Organization

The report is organized in many chapters. Chapter 1 gives an overview of the project. Chapter 2 illustrates the existing similar projects. Chapter 3 defines the project's requirements and what is needed to do the work. Chapter 4 designs the interfaces of the system. After collecting the information and creating the interfaces, that means everything is done to implement the system and write the implementation process with testing in Chapter 5. Chapter 6 writing the evaluation.

1.7 Conclusion

The processing of large customer reviews for opinion analysis remains challenging for traditional sentiment analysis methods. The project creates an AI-based sentiment analysis system through NLP and machine learning for review classification and insight extraction. The chapter starts by defining the issue while describing our goals and presenting information about the project framework and key participants and value.

Chapter 2

2.1 Project domain

Sentiment Analysis in the content creators and business analytics sectors forms the core focus of the project which operates within Artificial Intelligence (AI), Natural Language Processing (NLP) and Machine Learning (ML) domains. The system utilizes modern AI methods to examine customer comments and generate significant insights from them.

The project organizes opinions through sentiment classification to detect positive and negative and neutral statements while ABSA tracks product features in reviews. The system's goal is to improve organizational decisions by processing unstructured customer feedback into useful information that drives better satisfaction levels and product advancement initiatives.

2.2 Methodology

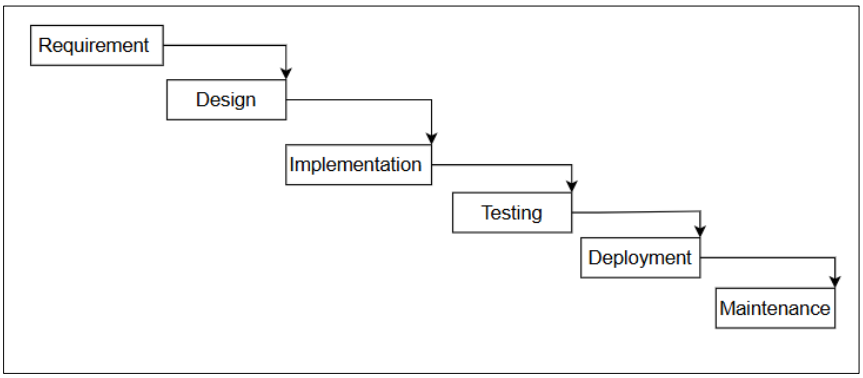


Figure 2-1: Waterfall Model

The project development process adopts waterfall methodology which represents a traditional method that uses linear sequential development (Conrad, 2011). The methodology serves our project well because it adapts to requirements stability which makes it an appropriate tool for development. Under the waterfall model the phases must complete their tasks independently before proceeding to the next stage while avoiding simultaneous phases and frequent requirement modifications. This approach defines a structured system building framework which provides efficient guidelines to develop the system. The project application of the waterfall methodology gets detailed in the subsequent section.

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1.RequirementAnalysis: Users need their requirements analyzed together with functionality and technical specifications.

2. Design: Construct the application architecture and user interface together with the database structure according to established requirements.

3. Implementation: Code the app according to the finalized design and integrate algorithms and database functionality.

4. Testing: Rigorously tests the app for functionality, usability, and performance.

5. Deployment: Package and release the app on intended platforms.

6. Maintenance: Address updates, bug fixes, and improvements based on user feedback.

2.3 Pre-project phase

This section will present the feasibility study for the project to ensure the feasibility of the implementation and availability of resources. This includes technical, operational, financial and schedule feasibility

2.3.1Technical feasibility

Technical feasibility assesses whether the project can be implemented using the available technology. It is divided into two main parts:

- Software requirement

Table 2-1: Technical feasibility

Category	Tools
Analysis, Design, and Team Collaboration Software	Microsoft Teams classic, Google Forms, Draw.io, Figma
Web Development Software	Python, Google Cloud AI Platform

- Hardware requirement

The hardware that will be needed is a computer with at least CPU core i7 RAM at least 32GB and access to internet .

2.3.2 Operational feasibility

To assess potential user satisfaction with the tool features, a survey was conducted and distributed among various user categories. A total of 7 content creators participated in the survey.

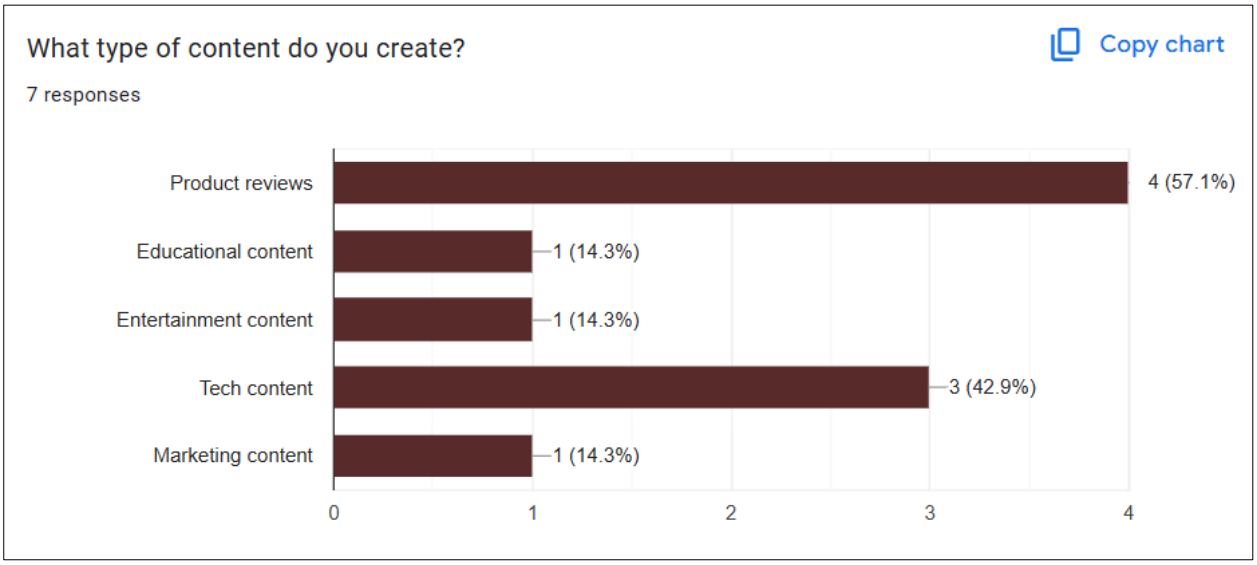


Figure 2-2: Product review creation percentage

This figure shows that **57.1%** create **product reviews**, making it the most common type. **Tech content** follows at **42.9%**, while **educational, entertainment, and marketing content** each account for **14.3%**.

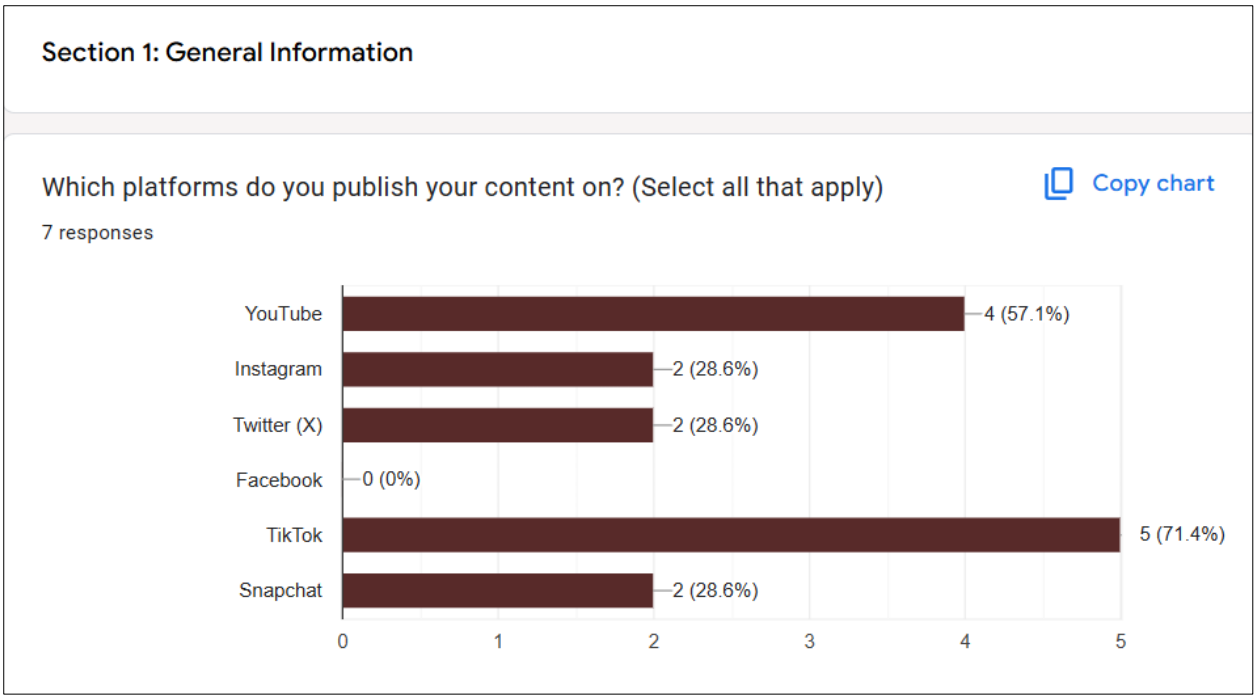


Figure 2-3: Most popular content platforms.

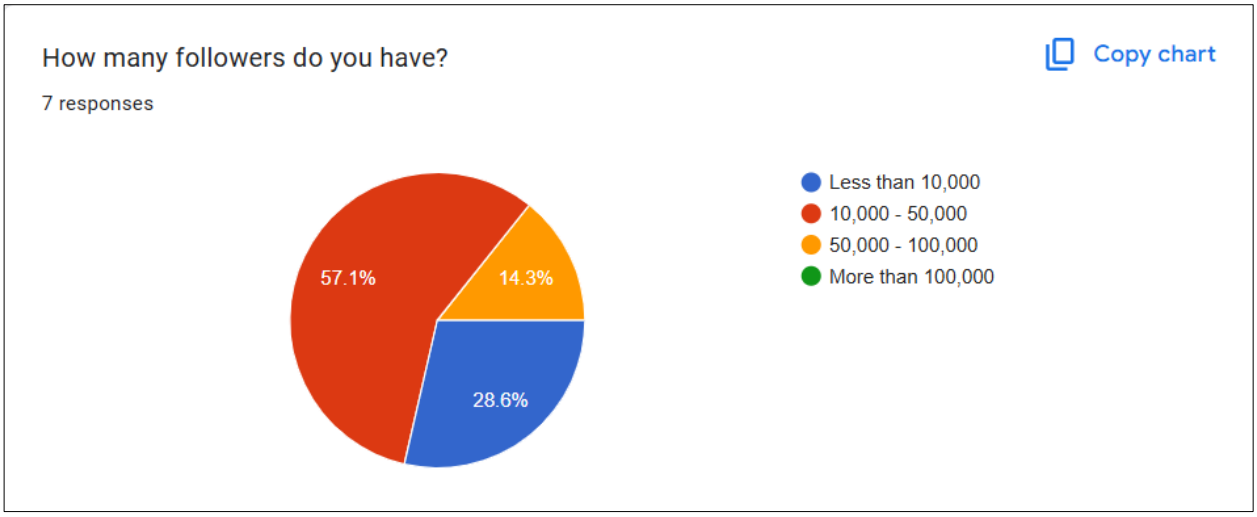


Figure 2-4: Dependence on product reviews for improvement

The chart shows that **TikTok (71.4%)** and **YouTube (57.1%) which** are the most popular platforms. **Instagram, Twitter (X), and Snapchat** each have **28.6%** usage, while **Facebook is not used at all (0%)**. This means a preference for video-focused platforms rather than traditional social media.

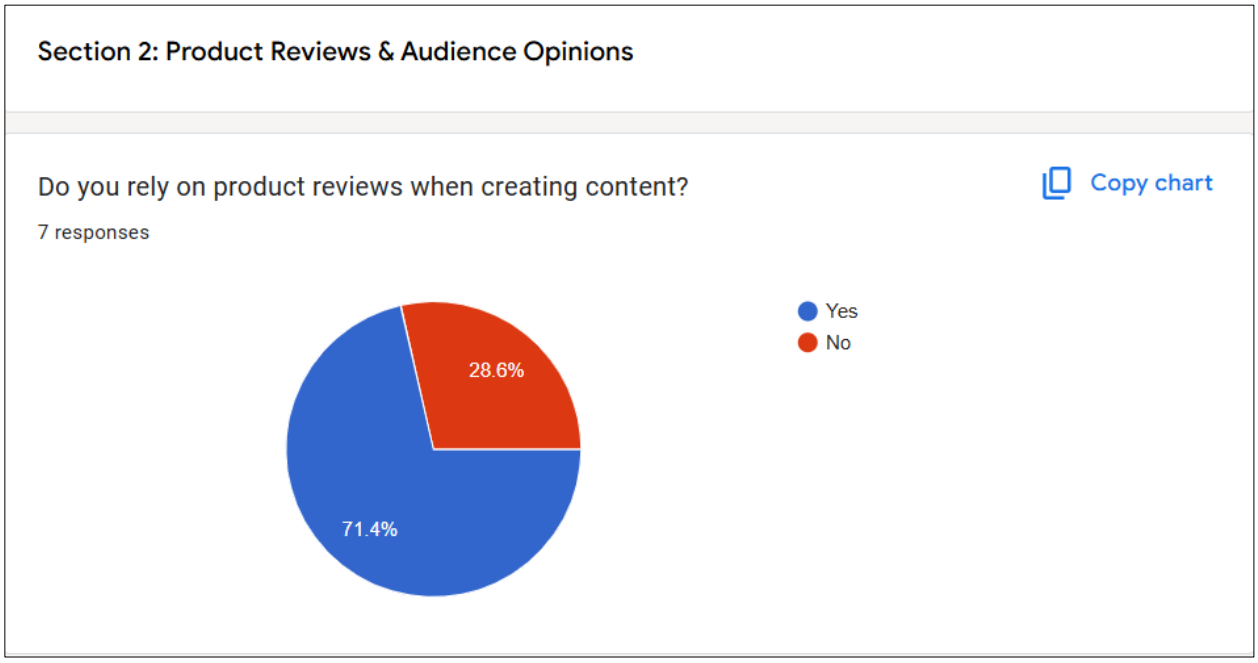


Figure 2-5: Common method for reading and analyzing comments

Here most of the responses rely on product reviews. Which means product reviews help them in improving their products.

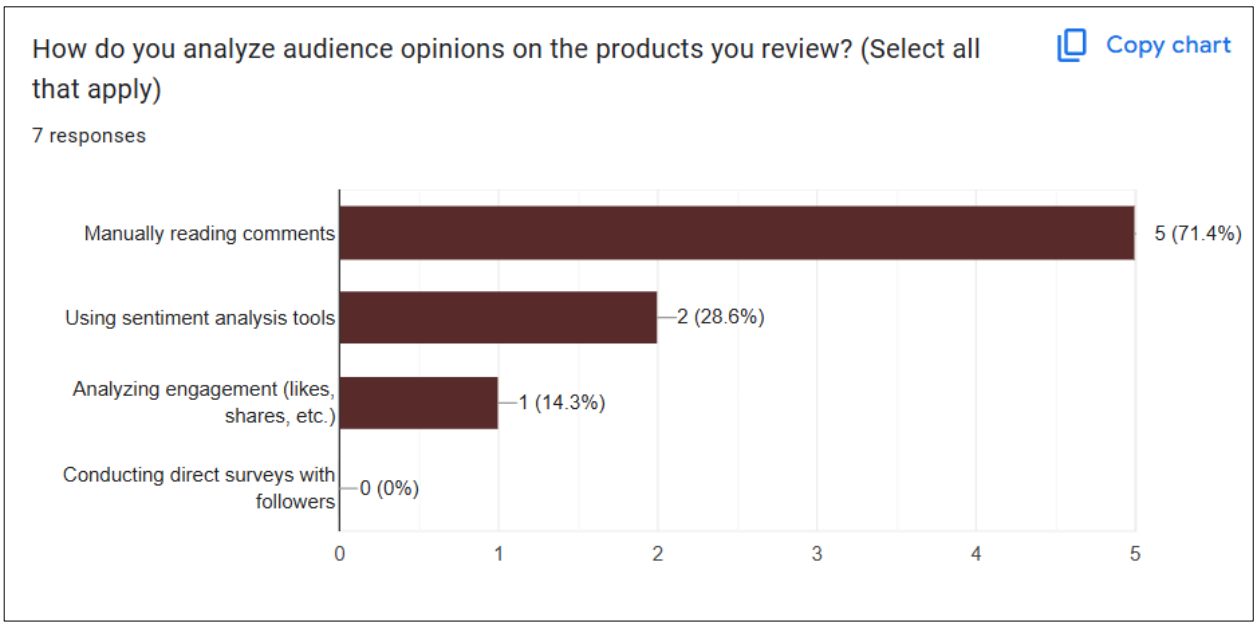


Figure 2-6: Major challenge of time consumption in review analysis

This figure shows that the common way is manually reading comments of the posts which is traditional.

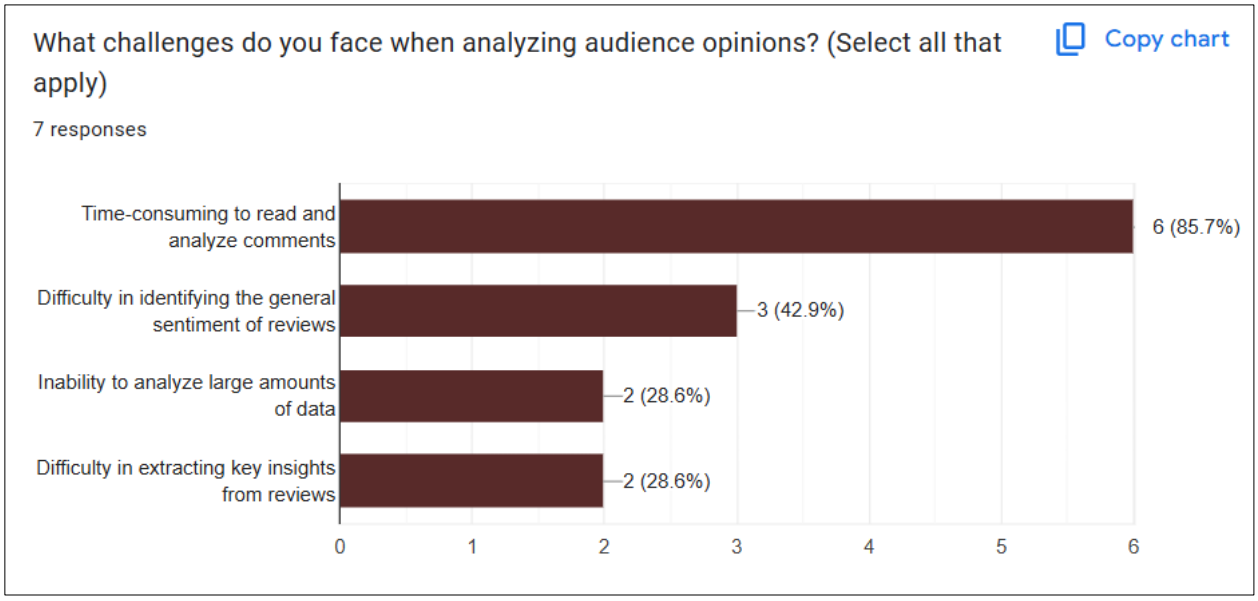


Figure 2-7: Perceived impact of product reviews on product enhancement

Here the responses shows the big challenge is time-consuming for analyzing the product reviews.

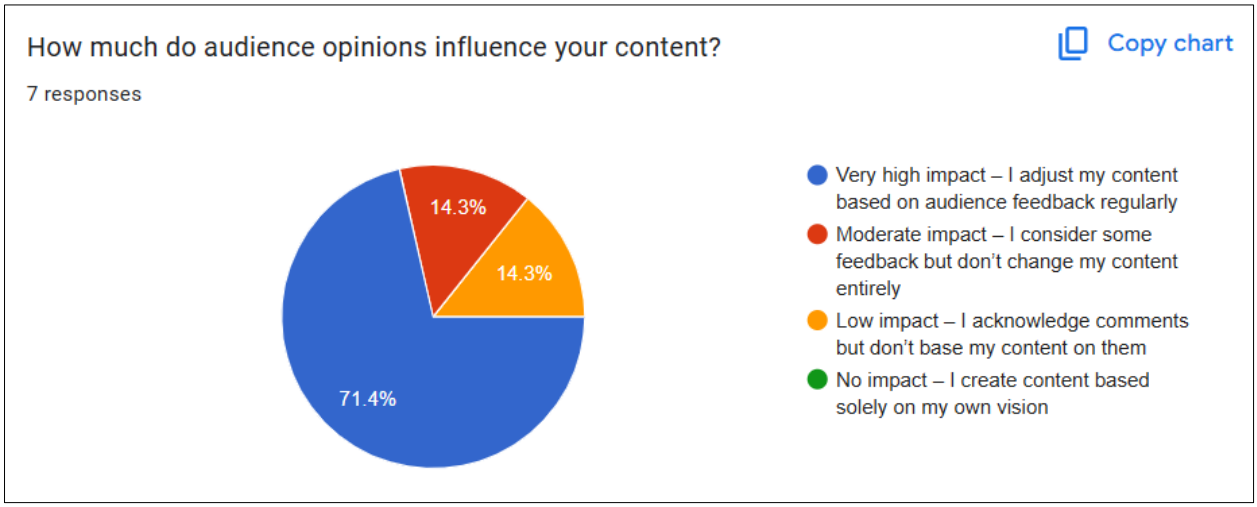


Figure 2-8: Agreement on the effectiveness of sentiment analysis tools

The responses show that 71.4% show that product reviews help organizations make their products better and better.

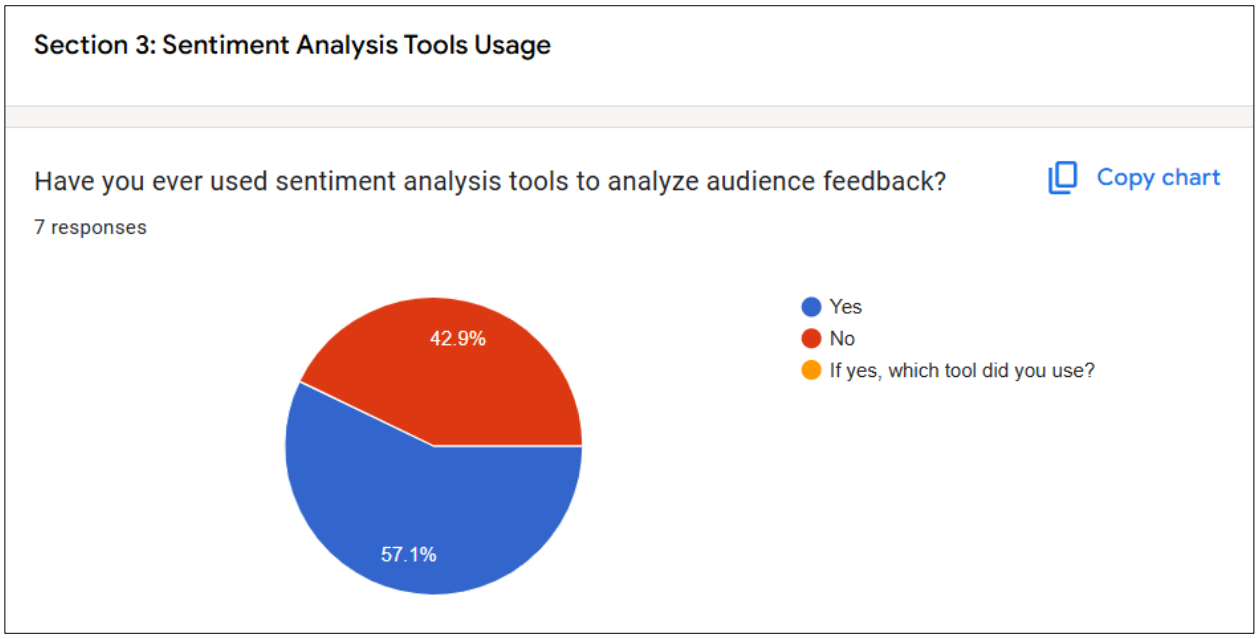


Figure 2-9: Preferred features for a sentiment analysis tool – high accuracy

it shows most of them use different sentiment tools to analyze the product reviews.

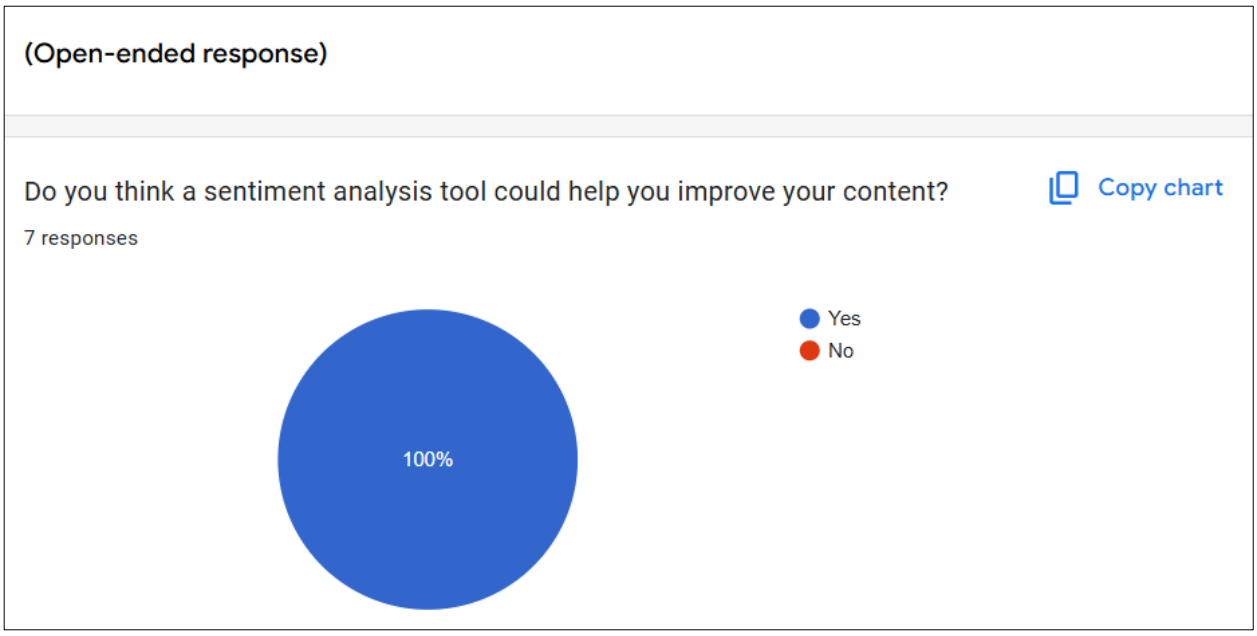


Figure 2-10: Preferred features for a sentiment analysis tool – sentiment classification

100% agree that sentiment tool help them in improving their content.

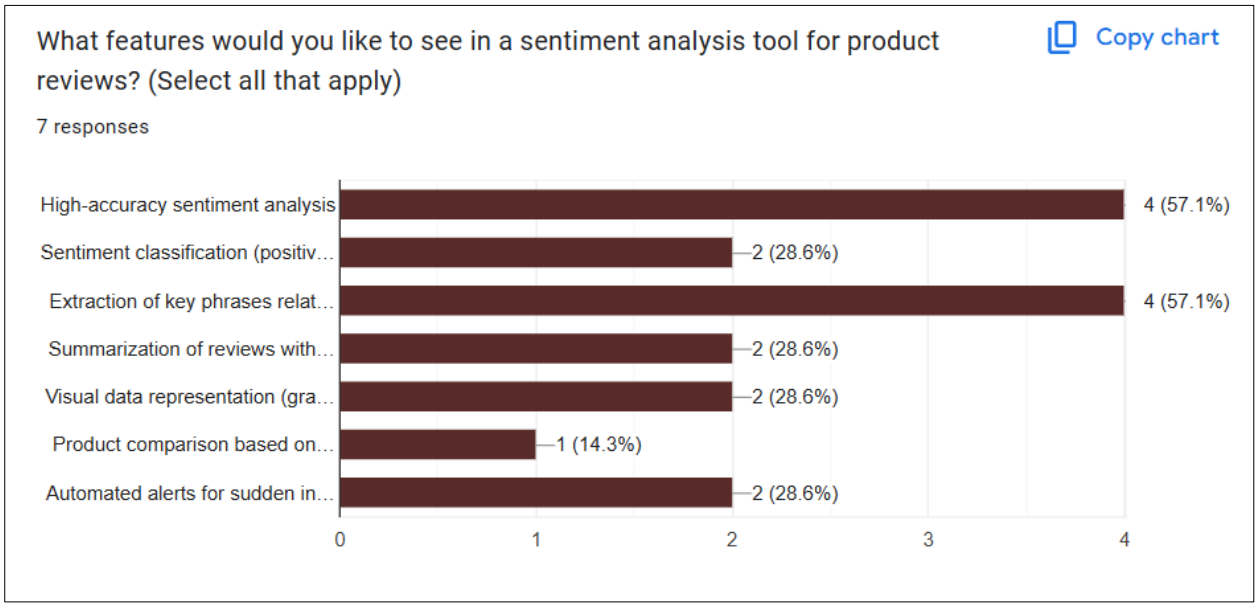


Figure 2-11: Preferred features for a sentiment analysis tool – key phrase extraction

The survey explored desired features for a sentiment analysis tool for product reviews with 7 responses:

1.High-accuracy sentiment analysis: 4 responses (57.1%)
2.Sentiment classification (positive/negative): 2 responses (28.6%)
3.Extraction of key phrases: 2 responses (28.6%)
4.Summarization of reviews: 4 responses (57.1%)
5.Visual data representation: 1 response (14.3%)
6.Product comparison: 2 responses (28.6%)
7.Automated alerts: 0 responses

Key preferences are for high accuracy, summarization, and sentiment classification.

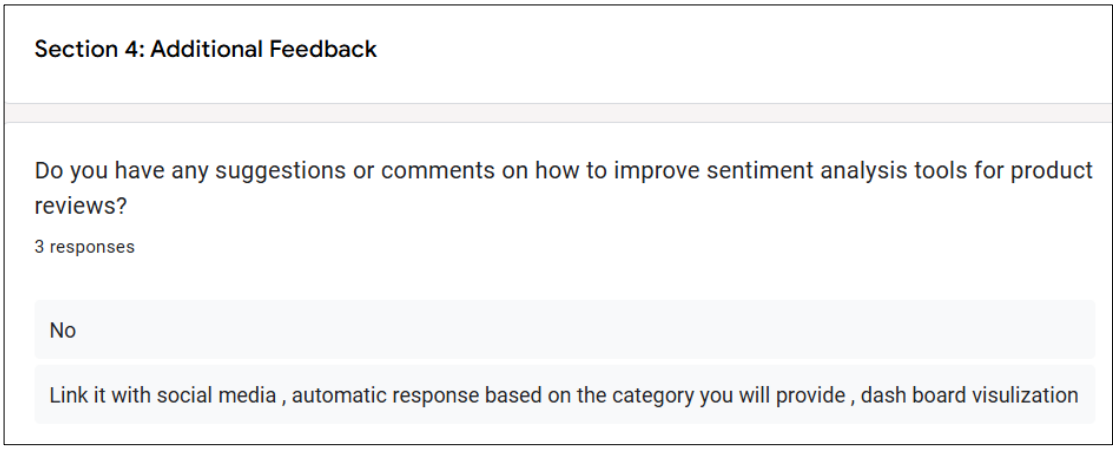


Figure 2-12: Additional Feedback on Sentiment Analysis Tools

2.3.3 Schedule feasibility

Table 2-2: Project Schedule

Phase	Task	Duration	Start Date	Finish Date
Requirement Analysis	Determine project idea	7 days	5/1/2025	12/1/2025
	Identify project objectives	6 days	12/1/2025	18/1/2025
	Identify project scope and stakeholders	5 days	18/1/2025	23/1/2025
	Describe the domain context	12 days	23/1/2025	4/2/2025
	Conduct feasibility study	15 days	4/2/2025	19/2/2025
	Gather project requirements	7 days	19/2/2025	26/2/2025
	Interim report	5 days	26/2/2025	1/3/2025
Design	Detail system design	20 days	1/3/2025	21/3/2025
	Prototype	15 days	21/3/2025	5/4/2025
Implementation	Coding	60 days	1/7/2025	1/9/2025
Testing	Test the app	8 days	1/9/2025	8/9/2025

2.3.4Financial feasibility

Open-source and freely available tools form the basis of our development plan to reduce project expenses. The expenses related to using cloud-based AI services for sentiment analysis depend on both text processing volume and the number of API requests made.

The estimated costs include:

Google Cloud Natural Language API: $1.00 per 1,000 characters for text analysis.

Entity Sentiment Analysis: $2.00 per 1,000 characters.

The per month fee for cloud storage amounts to $0.02 per GB.

The costs for using Compute Engine for AI processing range from $50 to $150 during each month.

The expenses vary according to pricing policies and the patterns of usage. Strategic monitoring combined with effective management plays a critical role in expense control for development as well as deployment.

2.3.5 User Survey

To further validate the project idea and support the implementation of the Sentiment Insights system, a user survey was conducted to explore actual consumer behavior concerning online reviews. The survey aimed to assess the extent to which users depend on customer opinions and how sentiment in product reviews influences their decisions. The following are key insights gathered from the survey :

Question: *Do you read customer reviews before purchasing a product?*

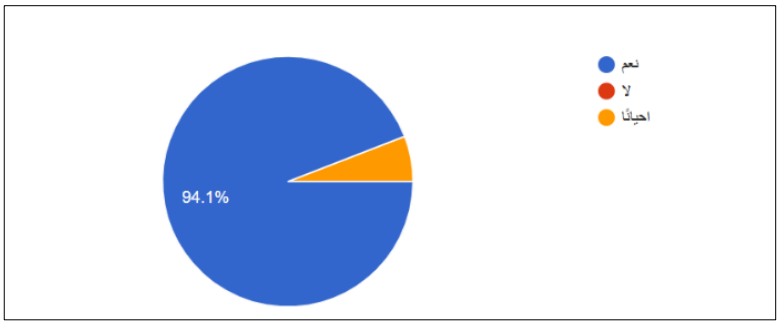


Figure 2-13: Importance of Others' Opinions Before Purchase

The survey results revealed that **94.1% of participants** read customer reviews before making a purchase decision, while only a very small percentage answered with "Sometimes" or "No."
This indicates the significant role that reviews play in influencing consumer behavior. It also highlights the need for tools that can **analyze and summarize customer opinions quickly and clearly**, which is exactly what the *Sentiment Insights* project aims to provide .

Question: *To what extent do customer opinions affect your decision to purchase a product?*

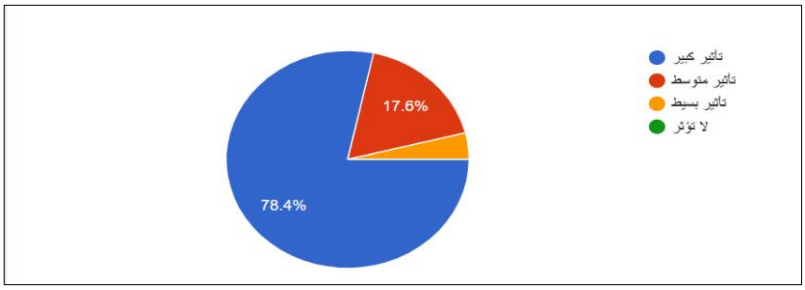


Figure 2-14: Influence of Others' Opinions on Purchase Decisions

These findings emphasize how heavily consumers rely on reviews when making purchasing choices. It reinforces the importance of tools like *Sentiment Insights* that help users quickly interpret and understand large volumes of customer feedback to make informed decisions.

Question: *Have you ever decided not to purchase a product because of a negative review?*

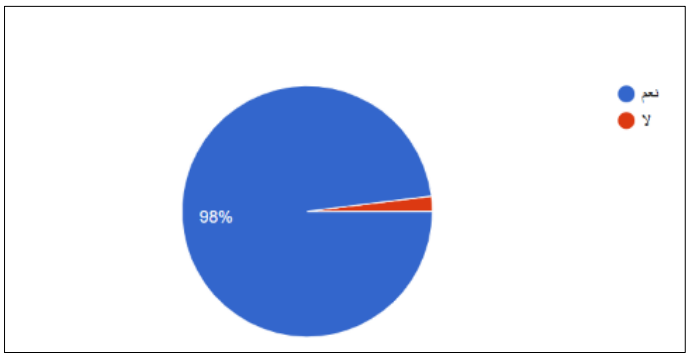


Figure 2-15: Reliance on Sentiment Analysis in Purchasing

An overwhelming **98% of participants** answered "Yes," confirming that negative reviews significantly influence consumer decisions. Only a very small minority responded "No."

This result further reinforces the crucial role of customer feedback in shaping purchasing behavior. It supports the purpose of the *Sentiment Insights* system, which aims to help users filter and understand sentiment in reviews, especially identifying negative trends that may affect their choices.

2.4 Related System

Sentiment analysis is a crucial tool in understanding emotions and opinions expressed in text. Traditionally, human analysts were required to manually interpret sentiments in written content, such as reviews, social media posts, and customer feedback. However, with advancements in artificial intelligence (AI) and natural language processing (NLP), automated sentiment analysis has become more accurate and efficient.Modern sentiment analysis tools leverage machine learning (ML) techniques, including deep learning models, to classify text into categories such as positive, negative, or neutral sentiments. These tools use techniques like word embeddings, recurrent neural networks (RNNs), and transformers to capture context and sentiment more effectively. The development of sentiment analysis has had a significant impact on fields such as market research, customer service, and social media monitoring, allowing businesses and content creators to gain insights into public opinion on a scale.

Analysis of Related Work:

As sentiment analysis continues to evolve, several AI-powered tools have been developed to analyze text and extract meaningful insights. These tools utilize natural language processing (NLP) and machine learning algorithms to classify emotions, opinions, and sentiments from various data sources. Below are key sentiment analysis tools that have been widely studied and implemented in real-world applications .

Monkey Learn is a cloud-based machine learning platform designed for sentiment analysis, text classification, and entity extraction. It allows users to train custom models without coding, making it accessible to both technical and non-technical users. The platform integrates with tools like Google Sheets, Excel, and Zapier, enabling seamless workflow automation (Monkey Learn, 2024). One of its biggest strengths is its flexibility—users can tailor models to fit specific industry needs while benefiting from real-time sentiment analysis. Its intuitive drag-and-drop interface simplifies the process, and the ability to process large datasets quickly makes it a valuable tool for businesses. However, its accuracy depends on the quality of training data, and its language support is somewhat limited compared to other sentiment analysis tools. Additionally, it struggles with detecting sarcasm and complex linguistic nuances, and some advanced features require a paid subscription (Monkey Learn, 2024) .

Meaning Cloud is a cloud-based sentiment analysis tool that uses natural language processing (NLP) and rule-based linguistic analysis to evaluate text. It supports multiple languages and can perform detailed sentiment analysis, including subjectivity detection, irony recognition, and aspect-based sentiment analysis (Meaning Cloud, 2024). This makes it especially useful for businesses that need in-depth insights into customer feedback, social media, or survey responses. The tool's API integration allows companies to embed sentiment analysis directly into their applications, enhancing automation and decision-making. One of its biggest advantages is its ability to analyze sentiment at a granular level, distinguishing between opinions and facts. However, the platform has a learning curve, as its advanced features may require a deeper understanding of NLP. Additionally, its processing speed can be slower when handling large datasets, and some of its more powerful capabilities are only available through a paid plan (Meaning Cloud, 2024) .

Repustate is a sentiment analysis platform that leverages machine learning and semantic analysis to extract insights from text, audio, and video content. Unlike many tools that rely on translation-based analysis, Repustate processes multiple languages natively, ensuring higher accuracy in sentiment detection (Repustate, 2024). It is particularly effective for real-time monitoring, capable of analyzing up to 1,000 comments per second. Additionally, its aspect-based analysis breaks down sentiments related to specific topics, offering more detailed insights. The platform is highly scalable, making it suitable for both small businesses and large enterprises, and it provides on-premises deployment options for organizations that require data security. One of its standout features is its ability to analyze media sentiment, including voice recordings and videos. However, its high level of accuracy comes with increased computational requirements, making it resource intensive. It also requires some customization to align with industry-specific needs, and its pricing can be on the higher side for businesses looking for a cost-effective solution (Repustate, 2024) .

Comparison with existing solutions:

The current sentiment analysis systems operated by Monkey Learn and Meaning Cloud with Repustate only perform basic sentiment categorization into positive negative and neutral categories without performing feature-based review analysis. Businesses struggle to determine product components which create satisfaction or dissatisfaction since traditional sentiment analysis systems fail to segment various product aspects. The failure of many tools to interpret sarcasm and implicit meaning causes them to incorrectly detect sentiment.

Aspect-Based Sentiment Analysis (ABSA) separates product features from review content to perform individual sentiment analysis on each feature. This solves the problems described above. Through the project businesses can obtain precise customer insights through their ability to identify which product aspects get rated positively or negatively by customers. Advanced algorithms within the system leverage Natural Language Processing (NLP) and Machine Learning (ML) features to provide enhanced sentiment analysis of complex document systems. The project design enables users to receive accurate real-time emotion detection as it automatically consolidates reviews through an interactive interface. The system works efficiently with existing business technology frameworks and provides affordability while delivering its services.

Table 2-3: *Competitive Analysis*

Ref No.	Tool Name	Objective	Methods/Models	Strengths	Challenges
[1]	MonkeyLearn	To provide customizable sentiment analysis and text classification without coding.	Machine learning-based sentiment analysis, text classification, and entity extraction. Uses NLP techniques and integrates with Google Sheets, Excel, and Zapier.	-Customizable models tailored to industry needs. - User-friendly drag-and-drop interface. - Real-time sentiment analysis. - Fast processing of large datasets.	- Accuracy depends on training data quality. - Limited language support. - Struggles with sarcasm and complex linguistic nuances. - Advanced features require a paid subscription.
[2]	MeaningCloud	To analyze text sentiment with detailed insights, including irony and subjectivity detection.	NLP and rule-based linguistic analysis. Provides aspect-based sentiment analysis and supports multiple languages.	- Multi-language support. - Aspect-based sentiment analysis. - API integration for automation. - Can distinguish between opinions and facts.	Learning curve for advanced features. - Slower processing speed for large datasets. - Some advanced features require a paid plan.
[3]	Repustate	To extract insights from text, audio, and video content using semantic analysis.	Machine learning and semantic analysis. Processes multiple languages natively. Supports real-time monitoring of up to 1,000 comments per second.	High accuracy with native language processing. - Effective for real-time monitoring. - Aspect-based analysis for detailed insights. - Can analyze media sentiment, including voice and video. - Scalable for enterprises and offers on-premise deployment.	- High computational requirements. - Requires customization for industry-specific needs. - Pricing may be high for budget-conscious businesses.

2.5 Business Model Canvas

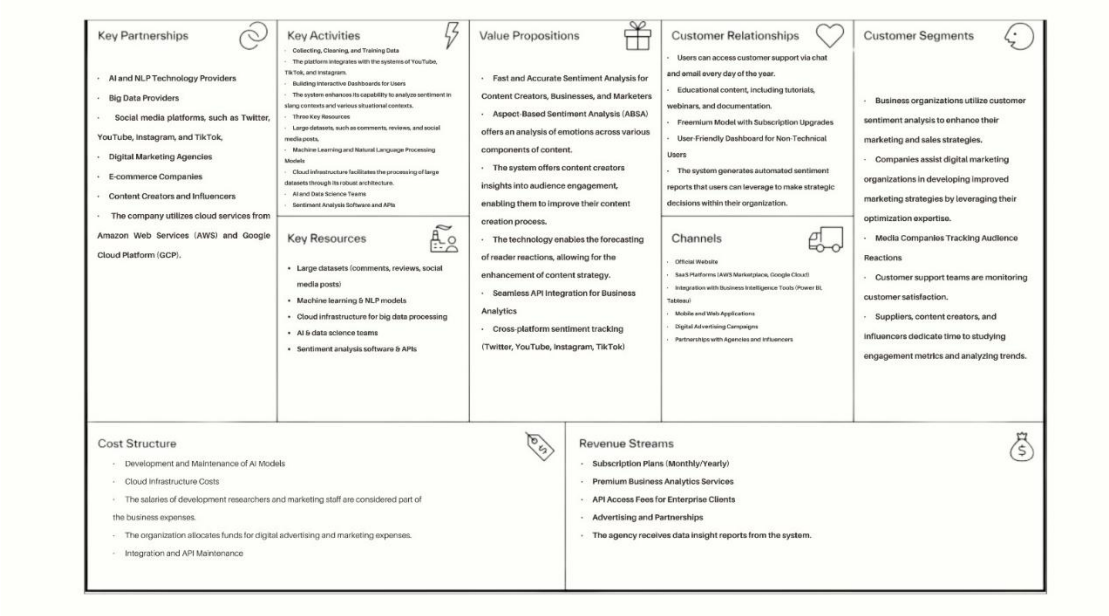


Figure 2-16: Business Model Canvas

2.6 Risk Analysis

In our world of unprecedented challenges and changes, risk analysis is an essential tool for identifying, assessing, and managing risks that may affect individuals or companies. It also aids in making accurate and informed decisions by evaluating the likelihood of risks occurring, which contributes to enhancing planning strategies and facilitating appropriate decision-making.

In our challenging and unprecedented world of change, risk analysis is an essential tool for identifying, assessing, and managing risks that may affect individuals or organizations. It aids in making accurate and informed decisions by evaluating the likelihood of risks occurring, which contributes to strengthening planning strategies and facilitating appropriate decision-making. Below, we outline some of the risks we aim to mitigate to ensure the success of our project and achieve our goals in the most effective manner:

1. Technical Risks:
    - Data Management and Preservation Issues: Data may be vulnerable to damage or loss due to issues within databases or deficiencies in backup procedures, which can compromise the accuracy of analyses and disrupt ongoing operations.

    - Internal Security Challenges: The system's architecture may contain internal vulnerabilities that could be exploited by malicious actors, potentially exposing it to cyberattacks and data breaches.

- System Failures and Malfunctions: The system may experience sudden failures that lead to service interruptions and delays in data processing, ultimately diminishing the user experience.

2. Security and Legal Risks:

- Violation of Privacy Laws: Collecting reviews may require the analysis of personal data, which can lead to legal complications.
- Potential Data Leakage or System Breach: If the system is compromised, the data may become vulnerable to leakage, posing significant security and privacy risks.

3. Ethical Hazards:

- Using the Tool for Unethical Purposes: Sentiment analysis may be employed to monitor individuals or negatively influence public opinion, raising ethical concerns and questions of social responsibility.

2.7 Conclusion

Sentiment analysis is a crucial tool that enables companies to gain a deeper understanding of audience sentiment and enhance communication channels between themselves and their customers. However, this system encounters several challenges, primarily two: first, its inability to accurately interpret emotions across various elements of text, and second, its limited adaptability to evolving language patterns. In this context, the ABSA project presents an innovative solution designed to provide businesses and content creators with exceptional precision, rapid processing, and an intuitive user interface. Our research in sentiment analysis has underscored its significance in supporting content development and business growth, while also evaluating the available tools and highlighting the unique weaknesses of each. Our business model incorporates essential elements that ensure integration and efficiency, and our project aims to revolutionize sentiment analysis by delivering a swift solution with enhanced accuracy and adaptable features to meet the demands of the expanding market.

Chapter 3

3.1 Introduction

In this chapter, we analyzed and specified the functional and non-functional requirements for. We outlined key system functionalities, including user authentication, file upload, sentiment processing, and result presentation through a user-friendly dashboard. Additionally, we presented essential diagrams—such as use case, activity, and sequence diagrams—to illustrate the interaction flow within the system. The design incorporates Aspect-Based Sentiment Analysis (ABSA) to deliver detailed sentiment insights.

3.2 Functional Requirements

1. User Authentication & Authorization

    1.1 User Registration

    - Users must be able to register using a valid email and password.
    - The system must validate the email format and make sure of the password's strength.

    1.2 User Login

    - Registered users must be able to log in using their email and password.
    - The system should provide proper authentication to prevent unauthorized logins.

    1.3 User Logout

    - Logged-in users must be able to securely log out from the application.

2. File Upload and Storage

    2.1 CSV File Upload

    - Users must be able to upload. CSV files containing data for sentiment analysis.
    - The system must validate the format and content of the uploaded file.

    2.2 File Storage

    - Uploaded files must be stored securely and associated with the correct user.
    - The system should maintain a history of uploads for each user.

3. Sentiment Analysis Processing

    3.1 Running Sentiment Analysis

- The application must send the uploaded CSV data to an AI model for sentiment analysis.
- The model should return:
  - ❖ Overall sentiment for each entry (positive, negative, neutral).
  - ❖ Aspect-Based Sentiment Analysis (ABSA) if the data supports it.
  - ❖ Feature Sentiment Analysis if features can be extracted.

3.2 Handling Unavailable Data

- o If ABSA or feature sentiment analysis is not available, the system should notify the user or simply omit it.

4. Results home screen

4.1 Analysis Viewing

- Users must be able to view the results of the sentiment analysis via a simple home screen.
- The home screen should include:
  - ❖ Sentiment labels (positive, negative, neutral).
  - ❖ ABSA results (if available).
  - ❖ Feature sentiment insights (if available).

4.2 Analysis History

- Users should be able to view past uploads and their analysis results.

3.2.1 Use Case Diagram

the Use Case Diagram shown in Figure 3.1 The diagram represents essential features which users can use within the application. The system produces a user journey which starts when users make either a fresh account or log into existing ones. A user who successfully authenticates through the system obtains entry to the primary dashboard page for interaction. Through the dashboard screen the user performs functions including uploading CSV files to send to AI processing and assessing the diagnosis and analysis reports. The user experiences an efficient CSV data management process through built-in actions that guide them throughout each operation. The way the user interacts with the AI system plays an essential role to create exact analysis after the successful data upload. The user can choose to log out which secures their session closure.

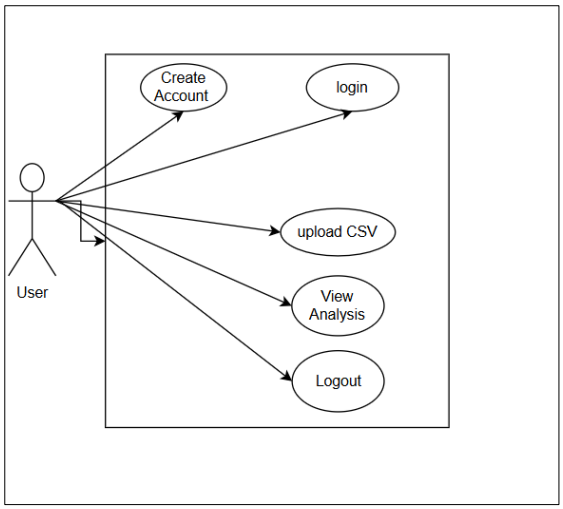


Figure 3-1: Use Case Diagram

3.2.2 Use Case Description

The use case description explains the different functions based on the system's requirements, shown in activity and use case diagrams. This section includes four detailed tables describing the main features of Sentiment Analysis™.

Table 3-1: Use Case Description

ID: 1	Use case name: Create an account with email.
Primary Actor	User.
Brief Description	The user creates a new account with their email in the Sentiment Analysis™ App.
Trigger	An unsigned user clicks on the sign-up button.
Precondition	User is not signed in.
Relationship	Association: User.
Flow of Events	The user enters their credentials in the sign-up form. The system sends the user a verification code via email. If the users enter the correct validation code, the system creates an account for the user and stores it in the database. The user is redirected to the login page.
Alternative flow	If the email is already in use, the system prompts the user to login instead. If the email is invalid, the user is prompted to enter a valid email. If the password is weak, the user is prompted to use a stronger password.
Post Condition	The user's account is created, and a successful screen is shown, prompting the user to sign in.
Exceptions	User's IP address is banned for some reason. Email is already in use. An internal server error.

Table 3-2: Use Case Description

ID: 2	Use case name: User uploads CFS file for analysis
Primary Actor	User.
Brief Description	The user uploads a CFS file to get a sentiment analysis shown.
Trigger	User clicks on the “Upload CFS” button.
Precondition	User is signed in.
Relationship	Association: User.
Flow of Events	The user signs in if they’re not already are. The user clicks on the “Upload CFS” button. The user then selects the desired file for analysis. The file is validated, and if it is a valid CFS it is saved and the path for the file is stored in the database under the user’s table. The user is shown a successful message. The system sends the file to the AI for sentiment analysis.
Alternative flow	The file’s extension is invalid. The file’s extension is valid, but the binary is not. The user is banned. Any internal server error.
Post Condition	The CFS file is uploaded, and the server awaits analysis from the AI
Exceptions	The file’s extension is invalid. The file’s extension is valid, but the binary is not. The user is banned. Any internal server error

Table 3-3 Use Case Description

ID: 3	Use case name: User displays sentiment analysis.
Primary Actor	User.
Brief Description	The user attempts to access the sentiment analysis associated with a given file.
Trigger	The user clicks on “Show Analysis” for a certain file.
Precondition	The user is signed in. A CFS file is already uploaded.
Relationship	User.
Flow of Events	User clicks on “Show Analysis” for a given file. The system checks if the analysis is ready, then shows it to the user.
Alternative flow	The analysis is still running, the user is shown an error message.
Post Condition	The user sees the analysis result.
Exceptions	The analysis is still running, the user is shown an error message. The user’s session expired. Any internal server error.

3.2.3 Activity Diagram

Figure 3.14 demonstrates the Sentiment Analysis system workflow through its activity diagram by showing the user interactions with the platform and AI module. The system starts with user authentication after which the system commences identification procedures. The user gains access to CSV upload after completing authentication successfully. The system transmits the uploaded file to the AI module right after the user completes the submission process. The AI component conducts the analysis while it extracts essential features and implements Aspect-Based Sentiment Analysis (ABSA). The analysis outcome is transmitted to the platform for storage until the user retrieves it through the system. The user views the analysis results after accessing them and completes their session by logging out. The flowchart visualizes all interaction phases through decision points while depicting the data movements throughout the system.

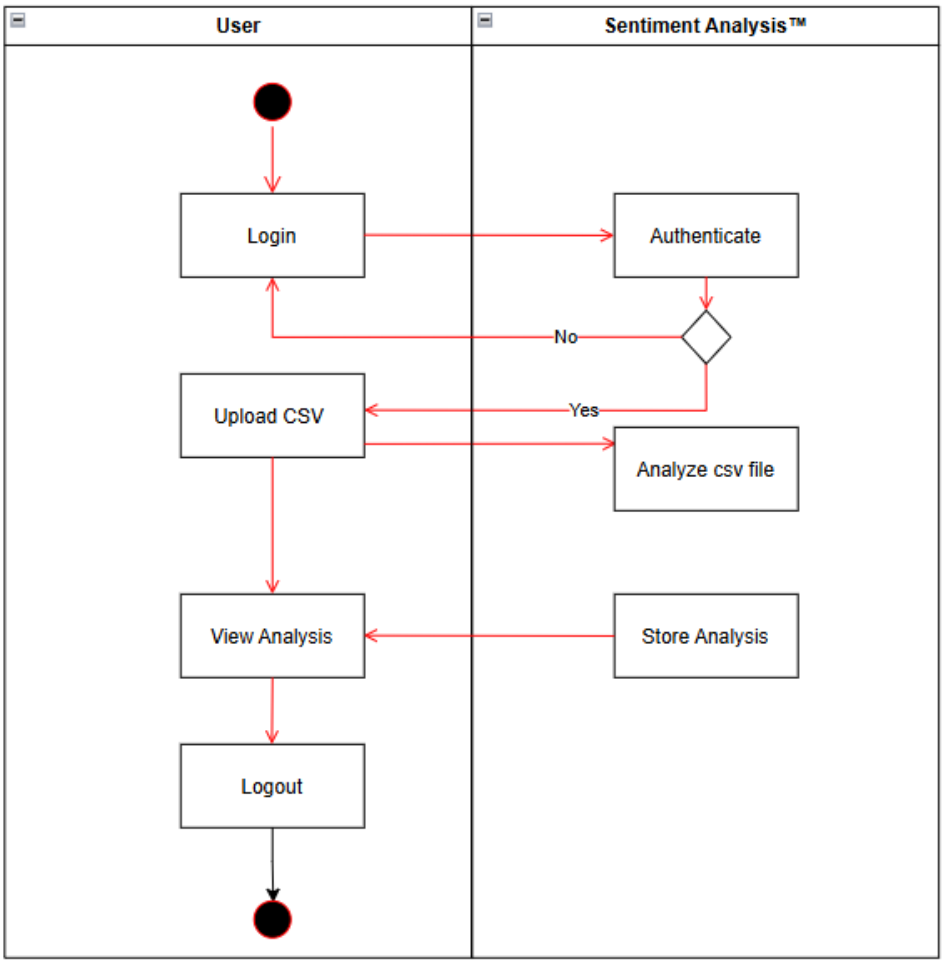


Figure 3.2 Activity Diagram

3.2.4 System Sequence Diagram

The system sequence diagrams shown in Figure 3.15 depict the interactive sequence between users and the Sentiment Analysis system while interacting with the database and AI module throughout two major use cases. The sequence starts once the user specifies their intention to either sign up or log in. After collecting user credentials, the system proceeds to verify the information against its database for authentication purposes. After verifying the user's credentials the system both provides interface access for the user and shows the main dashboard for display. The user submits Customer Feedback Statements (CSV) into the program during the second stage of operation. The system performs two functions by both storing the file within the database then distributing its content to the AI processing unit. The evaluation file goes to the AI for processing before the AI completes sentiment analysis and feature-based analysis then returns results to the system. The system saves analysis information to the database and displays those results on the analysis page that users reach through the dashboard. The diagrams present a clear depiction of the system operations that show how data travels through components as users engage with the system.

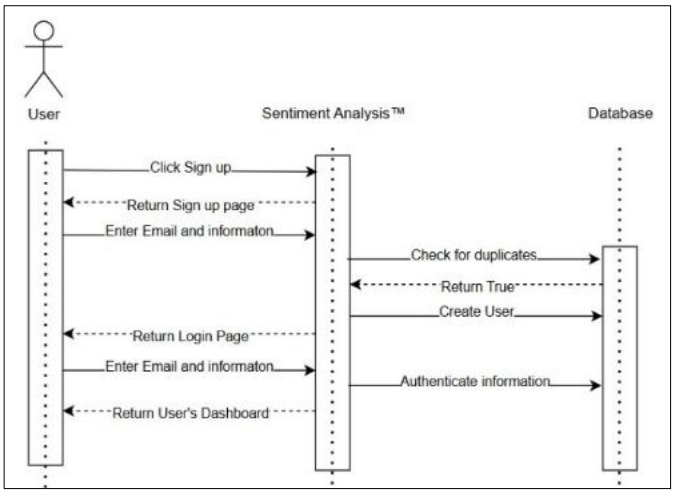


Figure 3-3: Sequence Diagram for User Login

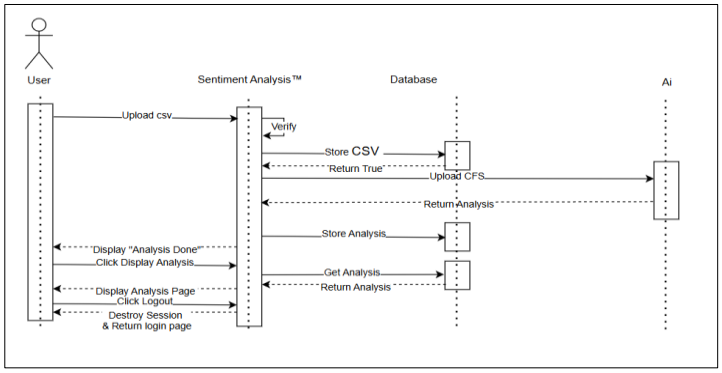


Figure 3-4: *Sequence Diagram for File Upload and Sentiment Analysis*

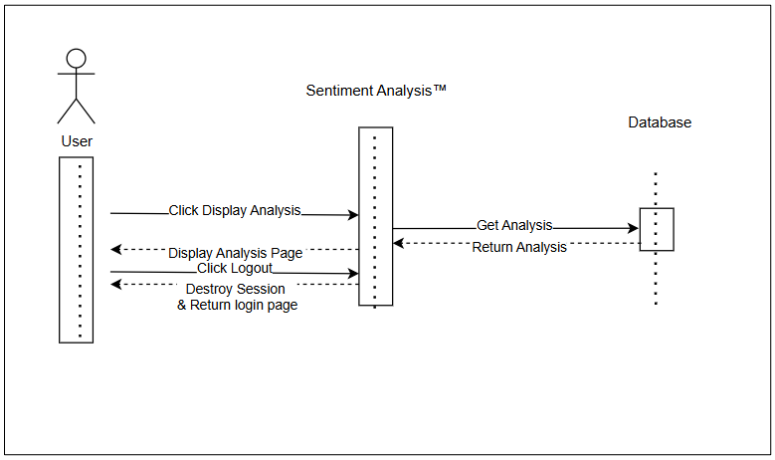


Figure 3-5: *Sequence Diagram for Viewing Analysis Results*

3.3 Non-Functional Requirements

1. Performance

- The app should respond quickly to user actions.
- Sentiment analysis results should be available within a reasonable time after upload.

2. Security

- Users' data should be protected.
- Only logged-in users should be able to access their own files and results.

3. Usability

- The system is easy to learn due to its user-friendly interface.
- It uses a consistent layout and terminology.
- Navigation is uniform across all screens for a smooth experience.

4. Compatibility

- The app should work on the common web browsers.
- It should also work well on both desktop and mobile devices.

3.4 Conclusion

This chapter focuses on requirements analysis and analysis models. into the sentiment analysis system by detailing the requirements acquisition process and presenting user interaction sequences. A comprehensive feedback system, leading to improved decision-making quality. analysis method ensures that the application delivers reliable and user-friendly performance while maintaining electronic security. Aspect-Based Sentiment Analysis (ABSA) enhances the accuracy of insights derived from user

Chapter 4

System Design

4.1 Introduction

This document presents an initial prototype of the user interface design for the "Sentiment Insights" application, which is designed to analyze customer sentiments toward products by processing textual data uploaded in CSV format.

The concept for this project was inspired by the increasing dependence of consumers on the opinions of others when making purchasing decisions. This observation prompted the creation of an intelligent tool designed to facilitate accurate and rapid sentiment analysis using advanced techniques**.**

4.2 Entity Relationship Diagram

Explanation of the Entity Relationship Diagram
The Entity Relationship Diagram (ERD) in Figure 4.1 illustrates how information is structured and related within the Sentiment Insights system.

User:

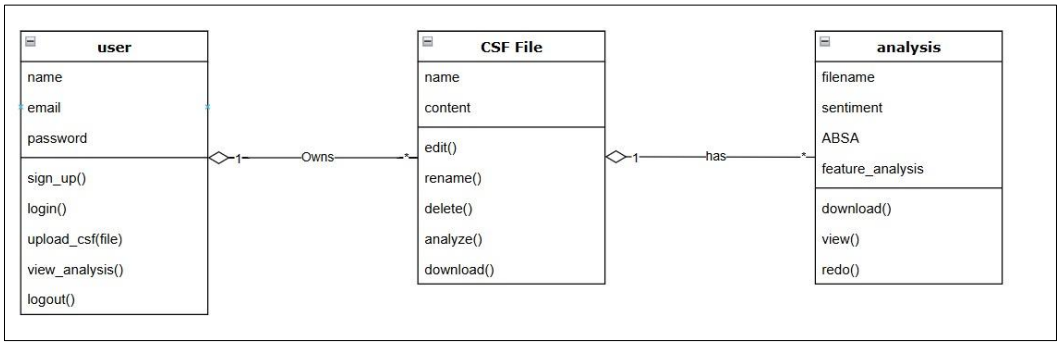
This entity represents the individual utilizing the platform. It stores essential attributes such as the user's name, email address, and password. A user can create a new account, log in to the system, upload CSV files for analysis, view the results of sentiment analysis, and log out upon completion. Each user has the capability to upload and manage multiple CSV files, establishing a one-to-many relationship with the CSV File entity.

CSV File:

This entity stores uploaded file data, including the file name and content. Users can edit the content of a file, rename it, delete it, trigger sentiment analysis, and download the processed file. Each file is associated with a single user and is linked to a corresponding analysis session, establishing a one-to-one relationship with the Analysis entity. ### Reason: Improved clarity and readability by refining sentence structure and enhancing vocabulary.

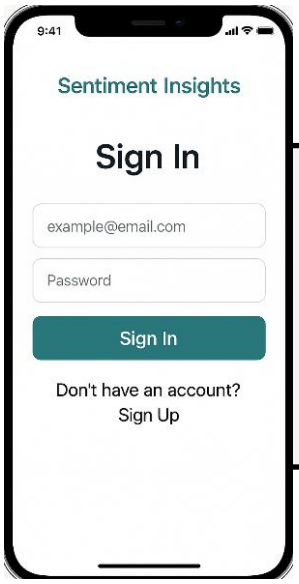
Analysis:

This entity captures the results of sentiment analysis performed on the CSV file. It includes the filename, sentiment results (Positive, Negative, Neutral), output from Aspect-Based Sentiment Analysis (ABSA), and feature-level sentiment analysis. Users can download the complete analysis, view results on the platform, or reanalyse the data if necessary. Each analysis is directly linked to a specific CSV file and provides detailed insights into the uploaded content.



Entity Relationship Diagram (ERD) in Figure 4.1

Prototype Design 4.3:



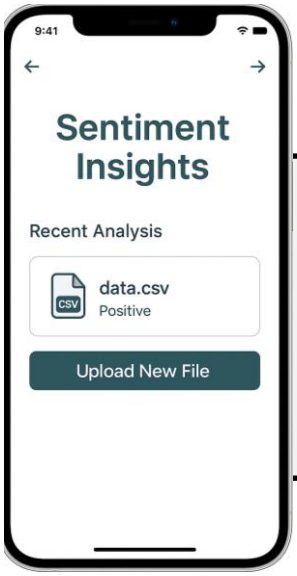
The Sign In interface allows users to log in using their email and password. It includes clear placeholder text to guide users while entering their credentials. A prominent “Sign In” button submits the information, and a “Sign Up” link directs new users to create an account. The layout is simple, intuitive, and optimized for mobile use.

Figure 4. 2 Sign In Interface

This screen displays the most recent sentiment analysis result, including the file name and its classification. Users can navigate through the app using the top arrows or the bottom. navigation bar for quick access to other features.



Figures 4.3 Home Interface



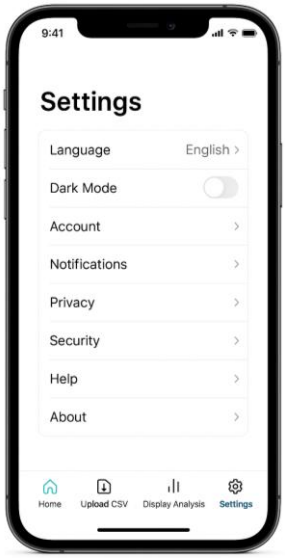
This screen allows users to view the most recent sentiment analysis and upload a new CSV file. It clearly displays the last analyzed file with its sentiment result, followed by a large, accessible “Upload New File” button. Navigation arrows at the top enable users to go back or forward between screens easily. The layout is user-friendly, modern, and mobile-optimized.

Figures 4.4 Upload New File Interface

This screen appears while the uploaded file is being analysed It shows a loading animation and a message indicating the analysis is in progress. Arrows allow users to navigate back or forward when needed.



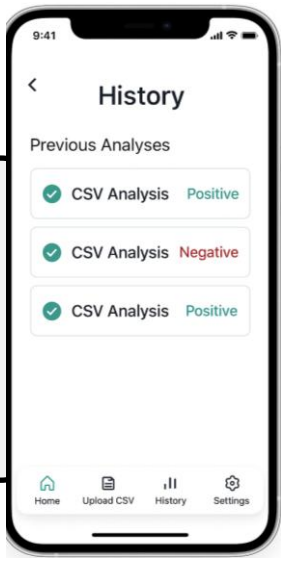
Figures 4.5 Processing Interface



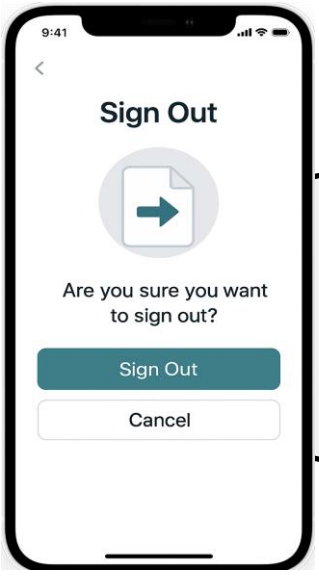
This screen provides users with access to various customization and control options, such as language selection, dark mode toggle, account settings, notifications, privacy, and security. It helps users manage their app experience efficiently and securely.

Figures 4.6 Settings Interface

This screen displays *a list of previous sentiment analysis results. Each entry shows the file type and its sentiment classification (e.g., Positive or Negative). The interface helps* users *easily review past analyses, with a clear layout and consistent color indicators.*



Figures 4.7 History Interfac



This screen confirms the user's intent to sign out of the app. It displays a clear message with two buttons: "Sign Out" to proceed and "Cancel" to return. The layout ensures clarity and prevents accidental sign-outs.

Figures 4.8 Sign Out Interface

4.4 conclusion

The user interface of *Sentiment Insights* was carefully designed to ensure a smooth, intuitive, and efficient user experience throughout the sentiment analysis process. From secure login to file upload, real-time processing, result history, and customizable settings, each screen plays a vital role in guiding users through the system with clarity and confidence. By prioritizing simplicity, consistency, and functionality, the interface empowers users to navigate the platform with ease—making sentiment analysis more accessible and impactful for everyday use.

Chapter 5: IMPLEMENTION

5.1 Introduction

This chapter describes the steps of making the system turn into a full-fledged mobile application, so that a user can experience the Sentiment Insights system in all of its features. To facilitate the app, a Python API was created that contains a collection of resources enabling the app to transmit textual data and obtain processed sentiment output. These resources are an essential part of the system architecture, to guarantee the smooth integration of the mobile application and the smart operations of processing.

Dataset Details Report 5.1.2

1. Dataset Type

The dataset that is used in the project is a CSV version of tweets and their respective sentiment labels. The records are single tweets with their metadata and classification fields.

Table 5-1: Dataset type

Column name	Data Type
tweet_date_created	Datetime
tweet_id	String/Number
tweet_text	String
language	String
sentiment	String
sentiment_score	Float

2.Dataset Size

The overall number of tweets in the dataset is about 101,689, which will be further divided into training and testing sets

Table 5-2: Dataset size

Number of Tweets	Subset
81,351 tweets	Training Set
20,338 tweets	Test Set
101,689 tweets	Total

*Train- test split ratio used: 80% training, 20% testing.

3. Sentiment Labels

There are four sentiment categories in the dataset. During preprocessing, these labels are transformed into numerical values in order to get the data ready for machine learning algorithms

*Table 5-3: **Sentiment categories***

Sentiment Label	Mapped Numeric Value
NEUTRAL	0
POSITIVE	1
NEGATIVE	2
MIXED	3

4. Tools used

The sentiment analysis system has been created with the help of some important tools and

technologies that have made it possible to process the data, train the model and integrate it to the back end.

Table 5-4: Tools used

Purpose	Tool / Library
Primary programming language for creating models and preprocessing data.	Python
The sentiment analysis API is deployed using a lightweight backend framework.	Flask
Utilized for loading, cleaning, and manipulating data	Pandas
Utilized for numerical operations in model computations and preprocessing.	NumPy
Library for label encoding, evaluation, and train-test splitting.	scikit-learn
Used in the construction and training of the neural network for sentiment classification.	TensorFlow / Keras

5. Data Preprocessing

The tweet dataset was subjected to a number of preprocessing procedures prior to machine-learning model training in order to guarantee data quality and enhance model accuracy.

Among the preprocessing were:

• URLs, mentions (user), and special characters are eliminated.

• To standardize the input format, all text is converted to lowercase.

• Eliminating unnecessary whitespace and duplicate tweet

• Tokenizing text by breaking it up into individual words.

• To get rid of non-informative words, use stop-word removal.

• Categorical sentiment labels are encoded into numerical form.

6.Machine learning mode

A supervised machine-learning neural network is used in the sentiment analysis model to categories tweets into four sentiment groups.

Model Characteristics:

• Dense Neural Network (DNN) is the type.

• Preprocessed and vectorized tweet text is the input.

• Four sentiment classes—Neutral, Positive, Negative, and Mixed—are produced.

• Adam is an optimizer for effective learning.

• Loss Function: For multi-class classification, categorical cross-entropy is a good choice.

• Activation functions include SoftMax for output layers and REL for hidden layers.

5.1.3 Software development tools

In this section, the focus is on the development environment, structures, databases, and other APIs that are factored into Sentiment Insights:

Android Studio: Android Studio has been selected as the Integrated Development Environment (IDE) due to the level of support it provides Flutter development and also the advanced debugging features. The enclosed Android emulator has been used to test, on a wide array of Android devices and Android Studio's fluent project arrangement and administration has also helped to set up a firmly arranged development workflow, which rendered it a perfect setting to create user interface elements and test the performance of the application across the stable development cycle.

Flutter framework: Flutter delivers development and execution of high-performance cross-platform mobile interfaces utilizing a common codebase. We selected it because it offers customizable user interfaces along, with fluid and seamless animations, fast rendering speed and speedy hot-reloading enabling us to apply interface modifications very rapidly throughout our development. Flutter enabled us to create contemporary, responsive, and visually uniform screens like the login page, home page, CSV upload interface, processing screen, results display and history page. These features resulted in Flutter being the best tool to use in the front-end of the Sentiment Insights application.

Firebase Authentication: Firebase authentication oversees the application's safe login procedures and user registration. It provides extensive support, such as email-password authentication, session management, and real-time identity verification. By limiting access to application features to only authorized users, this service improves system security and dependability.

Cloud Firestore: Cloud Firestore served as the main database for storing user-specific sentiment analysis history, timestamps, and associated metadata. Firestore's real-time capabilities and organized, document-based storage allowed the system to dynamically

retrieve and display prior analyses. Scalability, quick data synchronization, and smooth Flutter integration were guaranteed by Fire store.

Python Backend (API): The CSV files that were uploaded underwent sentiment analysis through a Python-driven backend. The platform employed machine learning and natural language processing techniques, utilizing libraries such as Pandas to analyze the datasets and generate sentiment categories (positive, negative and neutral). Using a lightweight framework like Flask or Fast API, the analysis model was made available as a RESTful API, allowing for smooth communication with the Flutter-based front-end application. The sentiment processing pipeline was kept accurate, modular, and effective thanks to this architecture.

5.1.4 Hardware Tools and Resources

This section outlines the hardware components utilized in developing Sentiment Insights. Multiple hardware devices were required to ensure both development and accurate testing.

• **Laptop:** Throughout the development phase a personal laptop was utilized to operate Android Studio, Flutter and the Python back-end setup. The laptop enabled code editing, debugging, emulator testing and running APIs.

• **Android emulator:** During the development phase every testing operation was performed using the Android emulator provided within Android Studio. This removed the necessity for a smartphone by providing a virtual device setting that enabled the testing of UI responsiveness, navigation flow, file-upload functionality and system performance in realistic scenarios. (Emulator, for Android).

• **Internet connection:** To access Firebase services, test API communication, and retrieve stored analysis history from the cloud, a reliable internet connection was necessary.

5.1.5 Functionality Implementation

This part describes the utilization of the Sentiment Insights apps functionalities across multiple pages. The app was developed using the Flutter framework within Android Studio supported by Firebase Authentication and Cloud Fire store. Each page was carefully designed to deliver an user experience ensuring intuitive navigation, proper data handling and easy access to the key features of the system. The purpose, structure, and behavior of each implemented page are explained in the ensuing subsections.

1-The Welcome Page

The welcome page is the initial interface that is shown once the application is opened and it is shown as Figure 5-1. This page gives the application logo and title placed in the middle of the screen in a simple and structured layout which gives a clear introduction to the system before the authentication process starts. The layout is based on a full background and placing the logo in the heart of the background to support the visual identity of the application. This page aims to provide an easy access point and redirect the user into the login/account creation screen. The welcome page lacks any interactive components as it acts as a transitional interface that is automatically closed or replaced by the succeeding authentication process.



Figure 5-1 Welcome Page

2- The Login Page

The safe entrance point of returning users appears in Figure 5-2 on the login page. The page also has well-marked email and password entry points, which are structured in the form of rounded-text boxes to ensure better readability and accessibility. There is a link that has an option of Forgot Password to help those who might have to retrieve their account. The primary button is the login button, which is placed in a prime position so that it can be easily accessed, and the secondary link is placed to the registration page, where they can get an account if not already having one. The presence of visual spacing and dark-themed interface helps with greater clarity and user experience. Attempts to log in are authenticate descriptive, easy to understand error messages shown in instances when the wrong credentials are typed.

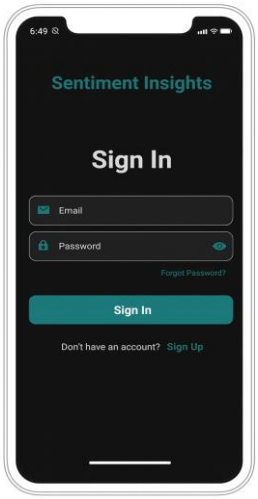


Figure 5-2 Login Page

3- The Sign-Up Page

Figure 5-3 shows the Sign-Up Page where new users can create an account. The interface also has some input fields such as full name, email address, phone number, password and confirm password with friendly icons to promote clarity and usability. A dynamic password-strength meter helps users to generate secure credentials by changing dynamically. The page has a structured vertical design with steady spacing and an obvious sign up button placed at the bottom. There is a shortcut link that allows users with an existing account to have direct navigation to the Login Page. In general, the design facilitates a successful onboarding process by making instructions clear and having first-rate error-checking mechanisms.

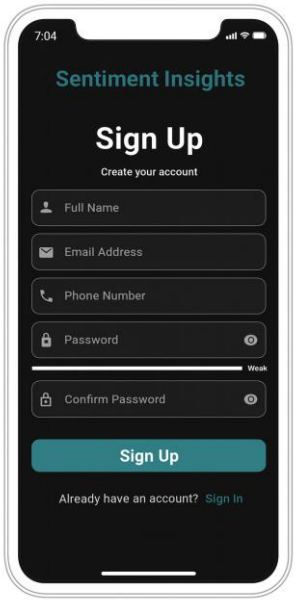


Figure 5- 3 The Sign-Up page

4- The Home Page

Figure 5-4 shows the Home Page, which is the primary dashboard to an authenticated user. The interface shows a plain welcome screen with the focus on the Analysis History section. In case no analyses have been done, the page shows a simple message that there is no data available. There is also an instructional section provided at the top of the page to assist users to know how to get or prepare a CSV file. This tutorial will be of special help to those users who do not specialize in dataset creation as it will help them get into the analysis process more readily. The bottom navigation bar will have links directly to the Home Page, CSV Upload Page, and the Settings Page, which will help users navigate easily within the application. It is a lightweight and user-friendly page that enables users to proceed with sentiment analysis workflow or navigate to the other features of the system easily.

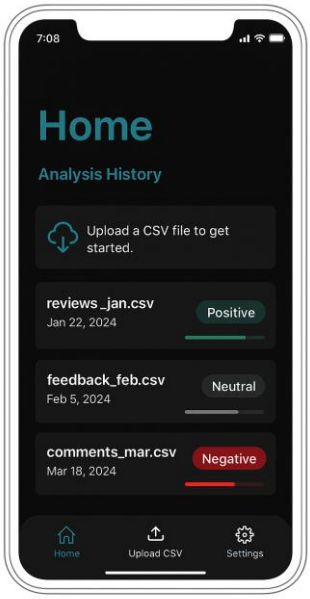


Figure 5- 4 Home Page

5- The Upload CSV Page

One of the main functionalities of the app is its CSV Upload Page as Figure 5-5 depicts. This interface enables one to upload a dataset to do sentiment analysis. On the upper part of the page, one can find a clear warning sign that the user should upload the file in English so that it can be processed as a proper model. The primary Upload New File button triggers the file selection feature of the device providing people with an opportunity to select a valid CSV file in Figure 5-6. Under this section, there is an optional field of commenting on the analysis, where users can leave additional comments on the analysis in Figure 5-7. Following the feedback on the examination committee on the challenge that some regular users might have in getting a CSV file, this field was extended to enable users to type short texts directly in the application to be analyzed. This will make the system more accessible, and the application will be able to be used without a CSV file. The page is clean and well designed with the help of bottom navigation bar. Validation tests are used to make sure that without a valid file being selected or appropriate text input inserted to be analyzed, a user may not proceed with the process.



Figure 5- 5 Upload CSV *Page*

6-The Analysis Results Page

The Results Page presents the outcomes produced by the sentiment analysis model, as illustrated in Figure 5-6. The page opens with a primary header labeled ‘Analysis Results,’ followed by a ‘Text Analysis’ section that provides a descriptive overview of the processed input. Further down, the system displays the overall sentiment outcome—such as positive, negative, or neutral—accompanied by an icon and visual cues that support quick and intuitive interpretation. A dedicated table titled ‘CSV Summary’ is also included, showcasing the processed rows along with their corresponding sentiment labels and confidence levels. This well-organized layout enables users to examine the analysis details in a structured, clear, and visually engaging manner.

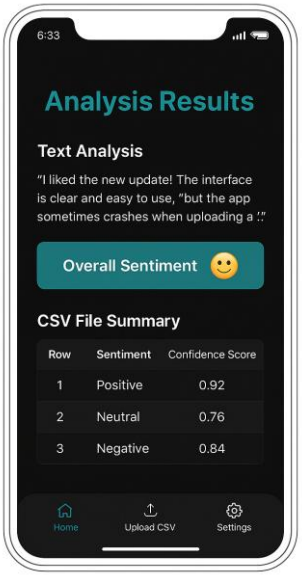


Figure 5- 6 Analysis Results Page

7- The Settings Page

The Settings Page allows users to customize and manage various aspects of the application, as shown in Figure 5-7. The interface includes multiple settings categories such as language, dark mode, account, privacy, security, help, and the 'About' section. Each category is presented as an individual card with a clear icon and a navigation arrow that directs the user to its corresponding subpage. From this page, users can enable the dark mode directly, as well as navigate to additional settings screens to view privacy details or manage account information. The page features a clean and organized design with well-spaced components and a professional layout, supported by a bottom navigation bar that provides quick access to other main sections of the application.

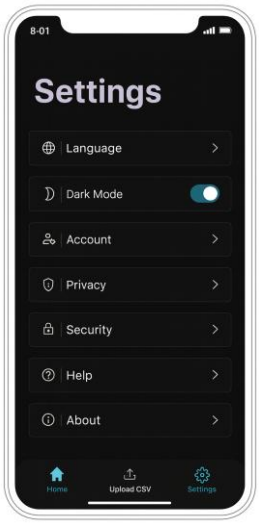


Figure 5- 7 Setting Page

5.1.6 Code Samples

This section presents representative code snippets from the core implementation modules. Each snippet is accompanied by a figure number and a brief explanation describing its function and role within the system.

1- Main upload and analysis method on the Upload Page

Figure 5-8 shows the main logic that the UploadScreen class uses to handle file uploads and start the sentiment analysis process. This implementation handles sending either a CSV file or an optional free-text comment to the backend sentiment analysis API. Firebase Firestore safely stores the processed results after a response is received. Then, through a dialog box, the system shows the user a visual summary, making sure they have a clear and quick view of the analysis result.

```
Future<void> uploadCSV() async {
  if (selectedFile == null) {
    ScaffoldMessenger.of(
      context,
    ).showSnackBar(const SnackBar(content: Text("No file selected")));
    return;
  }

  setState(() {...});

  try {
    var request = http.MultipartRequest(
      "POST",
      Uri.parse("http://192.168.100.81:5000/analyze"),
    );
    request.files.add(
      await http.MultipartFile.fromPath("file", selectedFile!.path),
    );
```

```
      var response = await request.send();
      if (response.statusCode == 200) {
        var respStr = await response.stream.bytesToString();
        setState(() {
          result = json.decode(respStr);
        });
        await uploadResultToFirestore(result!);
        showResultDialog(result!);


        setState(() {
          selectedFile!.delete();
          selectedFile=null;

        });
      } else {
        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(content: Text("Server Error: ${response.statusCode}")),
        );
      }
```

```
      );
    }
  } catch (e) {
    ScaffoldMessenger.of(
      context,
    ).showSnackBar(SnackBar(content: Text("Error: $e")));
  } finally {
    setState(() {
      isLoading = false; // إخفاء GIF بعد الانتهاء
    });
  }
}
```

Figure 5- 8 Upload and analysis code on the Upload Page

How it works:

1-Selecting the CSV File : The first thing the method does is see if a CSV file has been chosen. A notification message alerting the user that the necessary input is missing is shown if no file is chosen.
After that, the upload process stops right away and doesn't continue.

2-Sending the File to the API
- A multipart HTTP request is created to handle file transmission.
- The selected CSV file is attached to the multipart request.

- The request is then sent to the Flask backend endpoint responsible for processing and analyzing the file content.

3 - Receiving the Analysis Result:

- The application processes the returned payload if the server responds successfully (HTTP 200 OK).

- A usable Map structure is created by reading and decoding the response body from JSON format.

- The user interface (UI) can then update and present the processed sentiment output to the user by temporarily storing the parsed analysis result within the state of the screen**.**

4 - Storing the Result in Firebase Fire store:

- The result is saved inside the Fire store document of the authenticated user.

- This allows the user to view all previously generated analyses from the Home Page.

5- Displaying the Results: After Firestore is updated, a dialog box is displayed showing a clear and well-formatted summary of the analysis.

6. Cleanup: After processing, the locally uploaded file is deleted to free up device storage space.

7 - Sign-in authentication method on the Sign-In Page: The _submit() method in the Login Page is responsible for managing the complete user authentication workflow. It validates the email and password input fields, securely sends the credentials to Firebase Authentication, handles any error messages returned by the server, and finally redirects the user to the main application interface (HomeLayout) upon successful login.

```dart
Future<void> uploadText(String text) async {
  print("📌 Starting uploadText...");

  if (text.isEmpty) {
    print("⚠️ Text is empty!");
    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(content: Text("Please enter some text")),
    );
    return;
  }

  print("✉️ Sending text to server: $text");

  print("⏳ Loading started...");

  try {
    var response = await http.post(
      Uri.parse("http://192.168.100.81:5000/analyze"),
      headers: {"Content-Type": "application/json"},
      body: json.encode({"text": text}),
    );
```

```dart
print("📩 Server responded with status: ${response.statusCode}");
print("📩 Server raw response: ${response.body}");

if (response.statusCode == 200) {
  var resp = json.decode(response.body);
  print("✅ JSON decoded: $resp");

  setState(() {
    result = resp;
  });
  print("📌 Result saved to state.");

  print("📌 Showing result dialog...");
  showResultDialog(result!);
  await uploadResultToFirestore(result!);
} else {
  print("❌ Server Error: ${response.statusCode}");
  ScaffoldMessenger.of(context).showSnackBar(
    SnackBar(content: Text("Server Error: ${response.statusCode}")),
  );
}
```

```
      );
    }
  } catch (e) {
    print("❌ Exception: $e");
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text("Error: $e")),
    );
  } finally {
    print("■ Loading stopped.");
    print("🎉 uploadText Finished.");
  }
}
```

Figure 5- 9 Upload and analysis code on the Upload Page 2

How it works:

1-Input validation : The method first validates the form using _formKey.current State!. Validate(). It guarantees that the email field is not empty and has a valid email format. It also confirms that the password field is completed and has the required minimum length.
The method immediately stops execution and shows the pertinent error - message next to the affected field if any input field fails validation.

2- Starting the sign-in process : Once all input fields pass validation, the variable _isLoading is set to true, activating the loading indicator on the 'Login' button. This visual cue prevents the user from initiating multiple login attempts during the authentication process, ensuring a controlled and consistent user experience.

3-Sending credentials to Firebase Authentication:

-The FirebaseAuth.instance method is called. signInWithEmailAndPassword() using the user's password and shortened email address.

- Firebase Authentication checks the credentials against the backends' current user records.

- After a successful authentication, Home Layout smoothly replaces the current screen and takes the user straight to the main interface.

4-Handling authentication errors :

- The FirebaseAuth.instance method is called. signInWithEmailAndPassword() using the user's password and shortened email address.

- Firebase Authentication checks the credentials against the backend's current user records.
- After a successful authentication, Home Layout smoothly replaces the current screen and takes the user straight to the main interface.

5-Handling unexpected failures:

- Any unexpected errors that occur during the login attempt are captured by a general exception handler.
- When this happens, the method displays a standardized ‘Unexpected Error’ message, which is also localized through EasyLocalization, ensuring that the user receives a clear and accessible notification regardless of the app’s selected language.

6- Finishing the process and updating the UI : In the final block, _isLoading is set back to false (only if the widget is still mounted). This updates the UI, hides the loading spinner, and enables the “Sign In” button again so the user can retry if needed.

7- Main sign-in method on the Sign-In Page: The _submit() method orchestrates the core login flow on the Sign-In Page by validating the user’s credentials, delegating authentication to Firebase, routing successful logins to the main application layout, and providing clear feedback messages throughout the process.

```
Future<void> _submit() async {
  if (!_formKey.currentState!.validate()) return;

  setState(() => _isLoading = true);

  try {
    // Firebase Auth إنشاء المستخدم على
    UserCredential userCredential = await FirebaseAuth.instance
        .createUserWithEmailAndPassword(
      email: _emailCtrl.text.trim(),
      password: _passwordCtrl.text.trim(),
    );

    // تحديث اسم المستخدم
    await userCredential.user!.updateDisplayName(_nameCtrl.text.trim());
    await userCredential.user!.reload();

    // Firestore حفظ البيانات الأساسية في
    final user = userCredential.user;
    if (user != null) {
      await FirebaseFirestore.instance.collection("users").doc(user.uid).s
        "name": _nameCtrl.text.trim(),
        "email": _emailCtrl.text.trim(),
        "phone": _phoneCtrl.text.trim(),
```

```
      "uid": user.uid,
    });
  }

  if (!mounted) return;

  ScaffoldMessenger.of(
    context,
  ).showSnackBar(SnackBar(content: Text(tr('account_created'))));

  // الانتقال لصفحة رئيسية بعد التسجيل
  // Navigator.pushReplacementNamed(context, '/home');

} on FirebaseAuthException catch (e) {
  String errorMessage;

  switch (e.code) {
    case 'email-already-in-use':
      errorMessage = tr('email_already_in_use');
      break;
```

```
      case 'weak-password':
        errorMessage = tr('weak_password');
        break;
      default:
        errorMessage = '${tr('firebase_error')}: ${e.message}';
    }

    ScaffoldMessenger.of(
      context,
    ).showSnackBar(SnackBar(content: Text(errorMessage)));

    // الرجوع لشاشة تسجيل الدخول
    Navigator.of(context).pushAndRemoveUntil(
      MaterialPageRoute(builder: (context) => SignInScreen()),
          (route) => false,
    );
  } catch (e) {

    print(e.toString());
    ScaffoldMessenger.of(
      context,
```

Figure 5- 10 Main sign-in method on the Sign-In Page

How it works:

1- Validate the form: The method initially validates the required input fields, ensuring that both the email and password are provided and that the email format adheres to standard formatting rules.
2- Show loading state: Once the form is validated successfully, a loading indicator is activated to inform the user that the login process is in progress. This prevents repeated button presses and ensures a smooth authentication experience.
3- Firebase Authentication: The method sends the user's email and password to FirebaseAuth.signInWithEmailAndPassword to verify the provided credentials. Firebase checks whether the entered information matches an existing user - account in the backend.

2- Navigation on Successful Login: authentication is successful, the user is redirected to the HomeLayout main interface . A confirmation message is displayed through a SnackBar to indicate the successful login.

3- Smooth Error Handling: In cases where a FirebaseAuthException occurs—such as an incorrect password or a non-existent user—the method converts th error code into a clear, localized message that is easy for the user to understand. This message is then shown to the user to help guide them toward resolving the issue.

4- Loading State Deactivation: Regardless of whether the authentication succeeds or fails, the loading state is disabled at the end of the process.

The login button returns to its normal state, ensuring that the user interface - remains responsive.

5- Language and Dark Mode Toggle on the Settings Page

The Sentiment Insights application's Settings Page's fundamental implementation is demonstrated by this code sample. In addition to offering access to other sections like account, privacy, security, help, and about, the page lets users change important preferences like turning on dark mode and switching between Arabic and English. The implementation uses EasyLocalization to dynamically update the interface language and depends on a shared theme manager to maintain visual consistency. All settings options have a clean, consistent layout thanks to the use of reusable row components.

```
ProfileRowItem(
  title: tr("language"),
  leadingIcon: Icons.language,
  onTap: () async {
    // مثال لتغيير اللغة أثناء التشغيل
    final currentLocale = context.locale;
    if (currentLocale.languageCode == 'en') {
      await context.setLocale(const Locale('ar'));
    } else {
      await context.setLocale(const Locale('en'));
    }
    setState(() {});
  },
), ProfileRowItem

SizedBox(height: 2.h),

// الوضع الليلي 🔆
Container(
  decoration: BoxDecoration(
    color: Theme.of(context).brightness == Brightness.da
        ? const Color(0xff1E1E1E)
        : Colors.white,
```

```
    borderRadius: BorderRadius.circular(8),
    border: Border.all(
      color: Theme.of(context).brightness == Brightness.dark
          ? const Color(0xff333333)
          : const Color(0xffE5E7EB),
      width: 1.5,
    ), Border.all
  ), BoxDecoration
  padding: const EdgeInsets.symmetric(
    vertical: 4,
    horizontal: 20,
  ), EdgeInsets.symmetric
  width: double.infinity,
  child: Row(
    children: [
      Icon(
        Icons.mode_night_outlined,
        color: Theme.of(context).brightness == Brightness.dark
            ? Colors.white
            : Colors.black,
        size: 24,
      ), Icon
      SizedBox(width: 1.w),
```

```
        height: 3.h,
        width: 1.5,
        color: Theme.of(context).brightness == Brightness.dark
            ? Colors.grey[700]
            : Colors.grey[300],
      ), Container
      SizedBox(width: 1.5.w),
      Text(
        tr("dark_mode"),
        style: TextStyle(
          fontSize: 17.sp,
          color: Theme.of(context).brightness == Brightness.dark
              ? Colors.white
              : Colors.black,
        ), TextStyle
      ), Text
      const Spacer(),
      Switch(
        value: context.watch<ThemeManager>().isDarkMode,
        onChanged: (val) {
          context.read<ThemeManager>().toggleTheme();
        },
        trackOutlineColor:
        const WidgetStatePropertyAll(Colors.transparent),
        activeTrackColor: Colors.white24,
        activeColor: Colors.white,
        inactiveThumbColor: const Color(0xffE2E8F0),
        inactiveTrackColor: const Color(0xffE2E8F0),
        thumbIcon: MaterialStateProperty.resolveWith<Icon?>(
            (states) {
          return const Icon(
            Icons.circle,
            size: 25,
            color: Colors.white,
          ); Icon
        },
      ),
    ), Switch
  ],
```

Figure 5- 11 Language and Dark Mode Toggle on the Settings Page

How it works:

1- Language Toggle:
The program determines the current locale (English or Arabic) when the user taps the language row.
- When the interface is in English, it calls context.setLocale(const Locale('ar')); to switch to Arabic.
- The locale is returned to English if the interface is already in Arabic.
- The user interface is refreshed by a setState() call, guaranteeing that all translated labels instantly update to reflect the newly chosen language.

2- Dark Mode Switch:

ThemeManager is connected to a switch in the Dark Mode row. isDarkMode, guaranteeing that the user's desired appearance is reflected in the interface.
Toggling the switch updates the global theme between light and dark modes by triggering the toggleTheme() function.
Using Theme.of(context), container and text colors are chosen dynamically. brightness, guaranteeing that borders, labels, and icons in both themes remain aesthetically clear.
Regardless of the mode that is chosen, this implementation guarantees a smooth and uniform experience across all Sentiment Insights screens.

5.2 Testing

5.2.1 Introduction

The testing was a crucial stage of the Sentiment Insights system development. It was intended to confirm that all modules such as Flask backend, AI model, Flutter mobile application, and Firebase database worked as expected and delivered credible results. The testing was done to ensure that the flow of data was appropriate across the system, errors were managed effectively, and the system was performing as per expectations.

The development was based on a self-written Flask API instead of a third-party API, hence, full functional and integration testing was required. The testing process was broken down into two broad categories namely Unit Testing, that involved testing of individual components in the backend and Integration Testing involved testing how all modules interacted with one another after being integrated.

5.2.2 Unit Testing

Unit testing had been performed to test the different parts of the Sentiment Insights application on an individual basis prior to full integration. This step was necessary to make sure that every role performed its intended duties in the right way and in isolation. The main aim of the testing was to ensure that all modules such as authentication, data management, user settings and the AI sentiment model worked as anticipated, generated correct outputs and mishaps were dealt with properly.

All test cases were well-planned to reveal the possible problems at the early development stages. Valid and invalid inputs in the testing were used to determine how reliable the system is and how it would respond in various situations. The tables below provide a summary of the outcome of the unit testing of each key functionality.

Table 5-5: Unit Testing: Signup & Login

Test Description	Input Data	Expected Result	Pass/Fail
Register new user with valid credentials	Email: "user1@gmail.com", Password: "123456"	Account created successfully	Pass
Register with invalid email format	Email: "user1gmail.com"	Error message "Invalid email format" displayed	Pass
Login with correct credentials	Email: "user1@gmail.com", Password: "123456"	User logged in successfully and redirected to home page	Pass
Login with incorrect password	Email: "user1@gmail.com", Password: "wrong pass"	Error message "Incorrect password" displayed	Pass
Empty fields on login	Leave email and password empty	Display "Please fill in all required fields"	Pass
Password reset request	Email: "user1@gmail.com"	Password reset link sent successfully	Pass

Table 5.6— Unit Testing: Sentiment Analysis (AI Model)

Test Description	Input Data	Expected Result	Pass/Fail
Upload valid CSV file	File: "comments.csv" containing 50 text entries	File uploaded and processed successfully	Pass
Upload empty CSV file	File: "empty.csv"	Error message "Invalid or empty file" displayed	Pass
Upload unsupported format	File: "data.pdf"	Error message "Unsupported file format" displayed	Pass
Schedule daily analysis reminder	Frequency: Daily, Time: 9:00 AM	Reminder saved successfully and notification activated	Pass
Delete scheduled reminder	Select previously added reminder	Reminder deleted successfully	Pass

*Table 5-7: **-Unit Testing Edit Personal Information***

Test Description	Input Data	Expected Result	Pass/Fail
Edit username	Change "User1" → "Shahad A."	Profile name updated successfully and displayed in the app	Pass
Update email address	Change email from "old@gmail.com" → "new@gmail.com"	Email updated successfully and saved in Firebase	Pass
Change password	Enter old password: "123456", new password: "newpass123"	Password updated successfully and confirmation message displayed	Pass
Attempt to save with empty field	Leave email or name field blank	Error message "Please fill in all required fields" displayed	Pass
Delete user account	Click "Delete Account" button	Account deleted from Firebase Authentication and Firestore	Pass

*Table 5-8: **Unit Testing AI Chatbot (Sentiment Analysis Function)***

Test Description	Input Data	Expected Result	Pass/Fail
Analyze positive comment	"I really love this app!"	Output: "Positive"	Pass
Analyze negative comment	"This system is disappointing."	Output: "Negative"	Pass
Analyze neutral comment	"The app is okay, not bad."	Output: "Neutral"	Pass
Analyze large dataset	CSV with 100 mixed comments	All processed correctly with no crash	Pass
Send empty input	Input: ""	Error message "Invalid input text" displayed	Pass
Measure response time	File: 100 comments	Analysis completed in less than 5 seconds	Pass

5.2.3 Integration Testing

The unit test was followed by integration testing, which was conducted to confirm that all the modules of the Sentiment Insights system had worked properly as a unified integrated

application. The primary goal was to have adequate communication between the Flutter mobile interface, Flask backend API, and the AI sentiment model and the Firebase database.

This step aimed at the testing of the interaction between various components, data exchange and reaction in the real user conditions. In all the test cases, it was determined that the data sent by the mobile app was properly processed by the backend, analyzed with the help of the AI model, and stored and displayed with Firebase.

Table 5-9: Integration Testin

Test Case	Objectives	Test Case Description	Expected Result	Pass/Fail
IT-01	Verify API connection	Test the connection between Flutter app and Flask server using the *Dio* package.	The app successfully connects to the API and receives a valid response.	Pass
IT-02	Validate data upload and transmission	Upload CSV file from the mobile app to the Flask backend for analysis.	File transmitted successfully and prepared for processing.	Pass
IT-03	Confirm AI model integration	Ensure Flask backend sends uploaded text to the trained Transformer model for sentiment classification.	The model processes data and returns accurate sentiment predictions.	Pass
IT-04	Check JSON response format	Test that Flask returns analysis results in correct JSON structure.	JSON response contains sentiment label and confidence score.	Pass
IT-05	Display results on mobile interface	Verify that analyzed data from Flask is displayed correctly in Flutter UI.	Results appear instantly and match backend output.	Pass
IT-06	Verify Firebase Authentication	Test login, logout, and session validation during API use.	Only authenticated users can access the sentiment analysis function.	Pass
IT-07	Validate Fire store data storage	Check if sentiment results are saved correctly in Firebase Fire store.	Data stored accurately and can be retrieved later.	Pass
IT-08	Test multiple concurrent requests	Run simultaneous analysis requests from multiple users.	All requests processed successfully with minimal delay.	Pass
IT-09	Error handling during failed API call	Simulate invalid or lost network connection while sending a request.	App displays “Connection Error” message and retries successfully.	Pass
IT-10	Verify cross-platform compatibility	Test app performance on both Android and iOS simulators.	Application runs smoothly and displays consistent results.	Pass

Chapter 6

6.1 UAT or SU

This chapter focuses on the evaluation of the Sentiment Insights application in terms of the conclusions of the user acceptance testing (UAT), the key findings, the obstacles that were faced during the development and limitations imposed by existing system. The chapter also explains possible ways in which the application can be improved in the future to increase functionality, usability and overall effects of the application.

The assessment procedure shows the positive aspects of the system, makes some sharp comments on the areas which require improvement, and highlights the role of using the application in sentiment analysis and decision-making based on the data.

In order to perform the UAT, we selected a sample of users who comprised of university students, people well acquainted with mobile apps, and users interested in AI-based solutions. The participants were requested to engage with the application through logging in, providing text, loading CSV files, and sentiment analysis with single and bulk inputs. They were also asked to look over their results that were stored and leave comments on usability, speed, clarity of the interface and precision of the sentiment predictions.

The UAT was aimed at finding out whether the application fulfills its purpose of offering an efficient and accurate sentiment analysis but whether the user interface is intuitive and responsive. The response of the users was recorded and interpreted to establish the general trends, satisfaction rates and the needs to be improved.

6.2 Findings

Sentiment Insights application achieves its main goals and is well-performing in the main areas of its work. The combination of Flutter front end with Flask API and the AI model became stable and offered decent and precise sentiment analysis.

Flexibility in the form of analyzing single comments and bulk CSVs is also among the main assets of the system and was valued by the users as especially helpful. The attribute enables the user to handle large volumes of data with ease and therefore the application is applicable in academia, social media monitoring, and customer feedback analysis. The users were highly satisfied with the accuracy of the sentiment prediction saying that the artificial intelligence model always gave logical categorizations of positive, negative, and neutral remarks.

Besides, the Firebase Authentication system was tested with a positive rating on its reliability and security. The ease with which the participants could login and reset their passwords also enhanced the onboarding process as participants found it to be simple.

A notable observation is that the Flask backend was effective as it responded to API requests at a fast rate and was also reliable. The average time to analyze was within acceptable ranges, even with bigger datasets, which can be considered a good performance of the server.

The interface design was also simple and easy-to-use, and users were pleased with the entire design. They could find their way easily through pages, post files, analyze results and have access to personal information.

These results verified that Sentiment Insights is a practical and working tool that can be used to assist in real-life sentiment analysis effectively.

6.3 Limitations and Future Work

Although the system was quite successful, there were various limitations that were noted during the system development and evaluation.

6.3.1 Technical Limitations

The merger of Flutter and the Flask API was one of the major challenges that were experienced. To achieve stable communication, the request formatting and error reporting were to be handled carefully. In some instances, the mobile application was slow in handling larger files hence an additional optimization in data processing and background processing is required. Also, the work with AI models created issues in the dependency version and compatibility when deploying models. To have all the necessary Python packages, Transformer versions, and model files working synchronously, several modifications were to be made. The team was forced to reorganize components of the backend to make the API more stable and efficient. Firebase integration was also found to have technical challenges especially towards the synchronization of user information and stored results. To ensure updates instantly permeated devices, there was a need to restructure some sections of the database structure to ensure it was more reliable.

6.3.2 New Technology Adaptation

The use of the AI sentiment model necessitated the adjustment to the new technologies that were not familiar at first. The team was forced to acquire experience related to the fine-tuning of models, the administration of the API endpoints, and the ability to work with the JSON responses in general. The model, in some cases, gave inconsistent scores on ambiguous comments based on confidence which pointed at the difficulty in regulating the variability of AI. Another aspect of the model that needed work to improve the quality was the experimentation with the preprocessing methods, including cleaning the text, emoji management, and elimination of noise in the user comments. This is what was necessary to enhance the model and its consistency.

6.3.3 Time Management Limitations.

The other significant challenge was time management. The need to balance the project development of the mobile interface, incorporate the backend, and perform extensive testing was challenging. Testing phase, especially, took more time to detect problems associated with input verification, network failures, and unreliable responses.

To address this, the team has broken the development process into smaller tasks with definite deadlines and had a structured work process to prioritize important features. This strategy contributed to the continuation of the productivity and delivery of the application within the required timeframe, even with the strict time schedule.

Comprehensively, the issues that were experienced were useful learning experiences. They reinforced the team problem solving skills, enhanced technical skills, and ensured the relevance of thorough planning and agility.

6.4 Conclusion

To summarize, the Sentiment Insights application can be viewed as a valuable addition to sentiment analysis, as it brings mobile development, AI modeling, and back-end processing to one application. Even due to the challenges mentioned, the system was able to fulfill its major objectives, providing its users with effective and precise sentiment analysis experience.

The development experience has been invaluable as it involves a great deal of information about the integration of AI, data manipulation, and cross-platform interaction, which helped to make the product stronger and better organized.

In the future, the project has a huge growth potential. The application could be improved further in future, with making models more accurate, adding more languages, real-time streaming analysis, and increasing features of the visualization to enhance usability and impact of the application.

Altogether, Sentiment Insights provides a solid basis to further innovation and may be viewed as a useful tool when users want to have a more in-depth insight into the data and the general opinion.

References

1. Benjamin, Cheryl. “Sentiment Analysis on Product Reviews”, IEEE Xplore, Jan. 2025.
   https://ieeexplore.ieee.org
2. Rahman, A. et al. “Deep Learning-based Sentiment Analysis for Product Ranking”, IEEE Xplore, Oct. 2024.
   https://ieeexplore.ieee.org
3. “Sentiment Analysis of Customer Product Reviews Using Machine Learning”, IEEE Xplore, Jan. 2025.
   https://ieeexplore.ieee.org
4. Sentiment Analysis and Product Review Classification in E-commerce Platform, IEEE Xplore, 18 Jan. 2024.
   https://ieeexplore.ieee.org
5. “Sentiment Analysis of Amazon Product Reviews Based on NLP”, IEEE Xplore, Jan. 2025.
   https://ieeexplore.ieee.org
6. Pankaj, Mehta. “... A Review on Sentiment Analysis Methodologies, Practices, Applications” ResearchGate, 2025.
   https://www.researchgate.net
7. Liu, Bing. Sentiment Analysis and Opinion Mining. 23 Apr. 2025.
   https://www.cs.uic.edu/~liub/FBS/sentiment-analysis.html
8. “A Comprehensive Review on Sentiment Analysis”, Springer Link, 2008.
   https://link.springer.com
9. “Sentiment Analysis and Text Classification”, MonkeyLearn, 4 Sept. 2024.
   https://monkeylearn.com
10. “Natural Language Processing and Sentiment Analysis”, MeaningCloud, 28 Jul. 2024.
    https://www.meaningcloud.com
11. “AI-Powered Sentiment Analysis for Text, Audio, and Video”, Repustate, 2 Feb. 2024.
    https://www.repustate.com
12. Bo Pang and Lillian Lee. “Opinion Mining and Sentiment Analysis”, 2008.
    https://www.cs.cornell.edu/home/llee/opinion-mining-sentiment-analysis-survey.html
13. Cambria, E., et al. “New Avenues in Opinion Mining and Sentiment Analysis”, 2013.
    https://ieeexplore.ieee.org/document/6470648
14. Giachanou, A. and Crestani, F. “A Survey of Twitter Sentiment Analysis Methods”, 2016.
    https://dl.acm.org/doi/10.1145/2936206
15. Medhat, W., Hassan, A., and Korashy, H. “Sentiment Analysis Algorithms and Applications”, 2014.
    https://www.sciencedirect.com/science/article/pii/S1110866513000895

16. Kumara, A. and Sebastian, T.M., “Sentiment Analysis: A Perspective on its Past, Present, and Future”, 2012.
    🔗 https://link.springer.com/article/10.1007/s10462-011-9237-y
17. Ravi, K., and Ravi, V., “A Survey on Opinion Mining and Sentiment Analysis”, 3 Apr. 2015.
    🔗 https://www.sciencedirect.com/science/article/pii/S0957417415000906
18. Boiy, Eric. “Automatic Sentiment Analysis in Online Text”, n.d.
    🔗 https://www.researchgate.net/publication/221365718_Automatic_Sentiment_Analysis_in_Online_Text
19. “Functional vs Non-Functional Requirements”, GeeksforGeeks, 2023.
    🔗 https://www.geeksforgeeks.org
20. “UML Use Case Diagram Tutorial”, Lucidchart, n.d.
    🔗 https://www.lucidchart.com/pages/uml-use-case-diagram
21. “Use Case Diagram & Examples”, Inflectra, 2024.
    🔗 https://www.inflectra.com/articles/use-case-diagrams.aspx
22. “UML Activity Diagrams”, TutorialsPoint, 2023.
    🔗 https://www.tutorialspoint.com/uml/uml_activity_diagram.htm
23. “Introduction to UML Diagrams”, IBM Developer, n.d.